

Repression of PRMT activities sensitize homologous recombination-proficient ovarian and breast cancer cells to PARP inhibitor treatment

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eLife Assessment

This study presents a rather **valuable** finding that a combination of arginine methyltransferase inhibitors synergize with PARP inhibitors to kill ovarian and triple negative cancer cells. The evidence supporting the claims of the authors is **solid**, although some comments and elaborations in the main text would have enhanced the comprehension and clarity of the data. The work will be of interest to scientists working in the field of breast cancer.

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Abstract

Therapeutic epigenetic modulation is currently being evaluated in the clinic to sensitize homologous recombination (HR)-proficient tumors to PARP inhibitors. To broaden its clinical applicability and identify more effective combination strategies, we conducted a drug screen combining PARP inhibitors with 74 well-characterized epigenetic modulators targeting five major classes of epigenetic enzymes. Notably, both type I PRMT inhibitors and PRMT5 inhibitors scored highly in combination efficacy and clinical prioritization. PRMT inhibition significantly enhanced PARP inhibitor-induced DNA damage in HR-proficient ovarian and breast cancer cells. Mechanistically, PRMT suppression downregulates DNA damage repair genes and BRCA-associated pathways, while also modulating intrinsic innate immune

responses within cancer cells. Integrative analysis of large-scale genomic and functional datasets from TCGA and DepMap further supports PRMT1, PRMT4, and PRMT5 as promising therapeutic targets in oncology. Importantly, dual inhibition of PRMT1 and PRMT5 synergistically sensitizes tumors to PARP inhibitors. Collectively, our findings provide strong rationale for the clinical development of PRMT and PARP inhibitor combinations in HR-proficient ovarian and breast cancers.

Introduction

High-grade serous ovarian cancer (HGSOC) and triple-negative breast cancer (TNBC) share clinical and genomic characteristics, such as poor prognosis, homologous recombination (HR) deficiencies and potential immunoreactivity^{1–4}. There is an urgent unmet medical need to develop effective therapeutic strategies for these two diseases. Recently, limited targeted-therapy options, such as the Poly (ADP-ribose) polymerase inhibitors (PARPi), have been approved to treat HGSOC and TNBC in a frontline setting as maintenance therapies or in a recurrent/metastatic setting^{5–10}. However, the greatest clinical benefit from PARPi monotherapy has been mainly observed in HR-deficient tumors^{5–10}, only representing a minority of patients (less than 50% of patients even in HGSOC^{1,3}). Due to frequent defection of one or more DNA damage repair (DDR) pathways, increased replication stress, and higher endogenous DNA damage levels, cancer cells highly rely on the DDR pathways for survival compared with normal cells^{10–14}. During tumorigenesis, to solve these crises, certain DDR-related genes are pathologically upregulated in cancer cells. Epigenetic mechanisms represent a major strategy used by cancer cells to maintain or restore their abnormal need for DDR gene expression, resulting in a targetable vulnerability (**Figure 1A**). This “Achilles’ heel” can be targeted by epigenetic modulators (e.g., BETi^{15–17}, CDK7/12i^{18–20}, DNMTi^{21–23}, and HDACi^{24,25}) or small molecules that indirectly modulate DDR gene transcription (e.g., PI3KCAi^{26–29}, AKTi^{30,31}, mTORi³², and MEKi³³, **Figure 1B**). Several groups, including our own¹⁵, have developed an “induced PARPi sensitivity by epigenetic modulation” strategy^{9,10,15} to preferentially impair the transcription of certain DDR genes, thereby sensitizing PARPi intrinsic and acquired resistant tumors to PARPi treatment. For example, our previous discovery on the combination of BETi and PARPi¹⁵ is being evaluated in a Phase 1b/2 trial and achieving promising clinical results (NCT03901469)³⁴. Excitingly, due to cancer cells being extremely sensitive to the epigenetic repression of DDR genes compared with normal cells, a therapeutic window for these combinations has been observed during early clinical development^{23,28–31,34}. Although promising results have been reported in the clinic, the following two key questions still need to be addressed: First, among the clinically applicable epigenetic modulators, which epi-drug(s) can provide the strongest synergistic effects with PARPi? Second, can a BRCA-independent strategy be developed, given that most current strategies are mediated by repressing the expression of BRCA, which has been genomically mutated in acquired HR-proficient HGSOC/TNBC patients. Therefore, a highly effective and BRCA-independent strategy needs to be designed and evaluated.

Protein arginine methyltransferases (PRMTs) have emerged as attractive therapeutic targets in oncology owing to their crucial functions during tumorigenesis^{35–39}. Arginine methylation is a common post-translational modification of proteins, playing crucial functions in the regulation of transcription^{40–43}, splicing^{44–51}, DDR^{43,52–67}, and immune responses^{43,68,69}. By transferring a methyl group from S-adenosylmethionine, PRMTs catalyze the formation of arginine residues with monomethylarginine (MMA), asymmetrical dimethylarginine (aDMA), and symmetrical dimethylarginine (sDMA)^{35–39}. Each such methylation state affects protein function and interactions with other proteins or nucleic acids in different ways, leading to distinct functional consequences. Based on catalytic activities, nine PRMTs in mammals are divided into three groups. Type I (PRMT1/2/3/4/6/8) and type II (PRMT5/9) enzymes catalyze the formation of MMA as an intermediate before the establishment of aDMA and sDMA, respectively. PRMT7 is a type III enzyme that catalyzes only the formation of MMA. The activities of PRMT1 and PRMT5 are

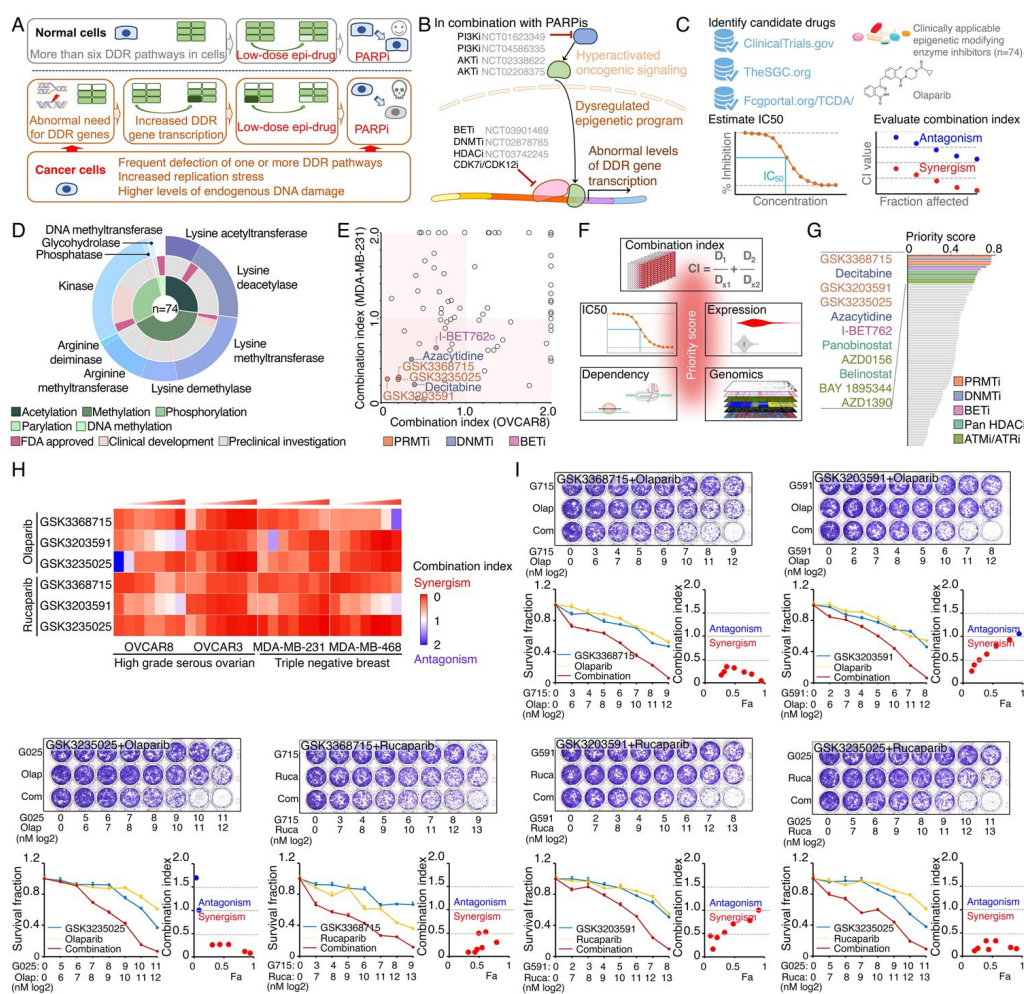


Figure 1.

Identification of synergistic actions between PRMTis and PARPi using a drug combination screen.

A. Illustration of the “induced PARPi sensitivity by epigenetic modulation” strategy for treatment of cancer patients with HR-proficient tumors. Due to frequent defection of one or more DDR pathways, increased replication stress, and higher endogenous DNA damage levels, cancer cells highly rely on the DDR pathways for survival compared with normal cells. During tumorigenesis, to solve these crises, certain DDR-related genes are pathologically upregulated in cancer cells. Epigenetic mechanisms represent a major strategy used by cancer cells to maintain or restore their abnormal need for DDR gene expression, resulting in a targetable vulnerability. This “Achilles’ heel” can be targeted by epigenetic modulators. **B.** Examples of the clinical trials based on the “induced PARPi sensitivity by epigenetic modulation” strategy. The combinations of PARPis with the epi-drugs or small molecules that indirectly modulate DDR gene transcription were evaluated in the clinic. **C.** Illustration the drug combination screen approach used in this study. **D.** The 74 epigenetic modifying enzyme inhibitors in the panel that was used for the drug combination screen were classified based on their mechanisms of action and target developmental levels. **E.** The combination index (CI) values were generated from a drug combination screen between olaparib and each epigenetic inhibitor in OVCAR8 and MDA-MB-231 cell lines. The CI quantitatively depicts synergism ($CI < 0.83$), additive effect ($CI = 0.83-1.2$), and antagonism ($CI > 1.2$). **F.** Five weighted features were collected and used to estimate the priority scores. **G.** The epigenetic inhibitors were ranked based on their priority scores, and the top prioritized inhibitors are highlighted. Color indicates mechanism of action. **H.** The heatmap shows average CI values between PARPis (olaparib or rucaparib) and PRMTis (GSK3368715, GSK3235025, or GSK3203591) in OVCAR8, OVCAR3, MDA-MB-231, and MDA-MB-468 cell lines. Color (red, synergism; blue, antagonism) intensity (light to dark) indicates increasing average CI values for each combination. **I.** An example of the sensitivity of cancer cells to PARPi alone, PRMTi alone, and a PARPi and PRMTi combination. Only results from OVCAR8 cells are shown here. The results from other cell lines are provided in Figure S1. For each treatment, the upper panel shows crystal violet staining of a colony formation assay; the lower left panel shows a quantified survival fraction; and the lower right panel shows the CI values. Fa, fraction affected.

responsible for generating most cellular aDMA and sDMA, respectively; therefore, they serve as the major type I and type II enzymes in cells, respectively^{37, 70}. Type I and II PRMTs share common substrates, with preferences for histones (such as H3R2 and H4R3)⁷¹ and proteins with RG/RGG motifs (such as RNA-binding proteins and splicing complex proteins)⁷². This substrate overlaps and competitive behavior leads to the phenomenon of substrate scavenging^{48, 73–75}. For example, PRMT1 and PRMT5 scavenge each other's substrates in cells⁷⁵.

Recently, potent and selective PRMT inhibitors (PRMTis) have been successfully developed, and more than ten PRMTis have been rapidly advanced into early clinical trials^{35–39}. Although relatively rare genomic alterations in the genes encoding PRMTs are observed in cancer, expression levels of PRMTs are remarkably upregulated in solid tumors, and elevated PRMT expression is significantly associated with poor clinical outcome. Therefore, PRMTs have been considered as attractive drug targets in oncology. Preclinical studies suggest that PRMT-targeting therapies may benefit tumors with MTAP deletion^{73, 76–79} or splicing dysfunction^{44–50, 80}. Although PRMTis provide a new avenue for cancer treatment, translational challenges need to be addressed for their clinical application. The PRMTi targetable “Achilles’ heels” (MTAP deletion and splicing dysfunction) are only observed in a small fraction of cancers, which restricts the application of a PRMTi as a clinical monotherapy. Therefore, the development of a novel treatment strategy to expand the potential clinical applications is a key challenge for PRMTis in oncology. Most importantly, because recent clinical trial results indicated that PRMTi activity alone may be insufficient to manage patients^{81–88}, even in MTAP deletion setting, pharmaceutical companies have had to terminate these trials and “re-think” how to administer these drugs to patients. Therefore, combination approaches of low-dose PRMTi with other existing therapeutic drugs urgently need to be designed and evaluated in the clinic. Excitingly, recent studies have revealed promising synergistic effects between PRMTis and chemotherapy, as well as DDR targeting drugs^{89–92}.

Results

Identification of synergistic actions between PRMTis and PARPi using a drug combination screen

To expand clinical applications and identify more efficient combinations for the “induced PARPi sensitivity by epigenetic modulation” strategy, we performed a drug screen by combining PARPi with a set of epigenetic modifying enzyme inhibitors, which contain 74 well-characterized epigenetic modulators that target 5 major classes of epigenetic enzymatic actions, histone acetylation, histone methylation, histone phosphorylation (only kinases and phosphatases that modify histone were included), histone de-ADP-ribosylation, and DNA methylation. Among these inhibitors, 7 are FDA-approved drugs in oncology, and 14 and 54 are in clinical trials and preclinical development, respectively (**Figure 1C** and **1D** and Table S1). The BET inhibitor I-BET762, a previously identified epigenetic modulator that synergistically acts with PARPi¹⁵, was used as a positive control. Two cancer cell lines that harbor wild type BRCA1/2 and modestly respond to PARPi (OVCAR8, HGSOc; and MDA-MB-231, TNBC) were used for screening. After the IC₅₀s of each drug were estimated in these two lines, five gradient doses (serially constant two-fold ratio dilutions) were used for the drug combination screen. The combination index (CI) value of olaparib with each epigenetic inhibitor was calculated to evaluate the synergy of the combinations. In summary, synergistic effects with olaparib (medium CI value <0.83) were identified in 33 and 21 epigenetic inhibitors in OVCAR8 and MDA-MB-231, respectively, including 10 that were shared by the two lines (**Figure 1E** and Table S1). Next, to prioritize the clinical application potentials, we estimated a priority score for each combination by considering the CI and four additional features of each epigenetic inhibitor (single agent treatment efficacy, cancer dependency defined by genetic screen from the DepMap project, and expression dysregulation and genomic alterations from the TCGA project, **Figure 1F**). As expected, candidates previously

identified by us¹⁵ and other groups, such as BETi^{15–17}, DNMTi^{21, 22}, and HDACi^{24, 25}, were ranked at the top of the list (**Figure 1G** and Table S1). Notably, a group of PRMT inhibitors (type I PRMTs targeting GSK3368715, and PRMT5 targeting GSK3203591 and GSK3235025) exhibited high priority scores in our analysis. This observation was further confirmed in an expanded panel of HGSOC/TNBC cell lines that were treated with each of these PRMTis in combination with two FDA-approved PARPis, olaparib and rucaparib, independently (**Figure 1H**, 1I, and Figure S1).

PRMT1, PRMT4/CARM1, and PRMT5 are potential therapeutic targets in oncology

To comprehensively characterize PRMTs in cancers, we analyzed the expressions and genomic alterations of all the PRMT family members across the TCGA cohort from 33 cancer types (Table S2). An RNA-sequencing (RNA-seq) analysis indicated that although all the PRMT family members were detectable in cancers, five PRMTs (PRMT1/2/4/5/7) exhibited remarkably high expression levels (average FPKM >10, **Figure 2A**, Figure S2A and Table S3). We also characterized expressional differences between tumors and their corresponding normal adjacent tissues. A meta-analysis across 21 cancer types, which had sufficient numbers of normal controls ($n \geq 3$), revealed that the mRNA levels of most PRMTs, except for PRMT2/9, were significantly upregulated in tumor specimens compared with corresponding controls (**Figure 2B** and Figure S2B). This observation was further validated at protein levels in an independent sample cohort from the CPTAC project as assessed by a mass spectrometry analysis (**Figure 2C** and Figure S2C). Most importantly, a meta-analysis across 33 cancer types from the TCGA cohort demonstrated that higher expressional levels of PRMT1/3/4/5 were positively correlated with shorter patient survival times (**Figure 2D**). Next, we analyzed recurrent genomic alterations (focal somatic copy number alteration, recurrent mutation, and recurrent transcript fusion) of the PRMT family across the TCGA cohort, and no significantly recurrent events were identified at either the individual or pan cancer levels (Table S4). Finally, cancer cell-growth dependencies of PRMTs were analyzed in a large-scale CRISPR screening dataset generated by the DepMap project. Consistence with their increased expression levels and associations with poor clinical outcomes, PRMT1/4/5 showed cancer cell-growth dependencies (**Figure 2E** and Table S5). The analyses of these five features indicated that two type I PRMTs (PRMT1 and PRMT4/CARM1) and one type II PRMT (PRMT5) may serve as potential therapeutic targets in oncology (**Figure 2F**). Because a PRMT4-specific inhibitor, CARM1-IN-1, did not show significantly synergistic effects with PARPi (**Figure 1B** and Table S1), we hypothesized that the synergistic function of type I PRMT inhibitor GSK3368715 may be mediated by inhibiting PRMT1, which is the predominant enzyme among type I PRMTs (responsible for >90% activity)⁷⁰. Consequently, we used CRISPR/Cas9 as a genetic tool to specifically knock out PRMT1 and PRMT5 independently in OVCAR8 and MDA-MB-231 cells (**Figure 2G** and **2H**). Colony formation assays demonstrated that an acute knock out of PRMT1 or PRMT5 significantly increased the sensitivity levels of cancer cells to PARPi treatment, thereby phenocopying the results obtained using chemical PRMT inhibitors (**Figure 2I** and **2J**).

PRMT inhibition enhances PARPi treatment-induced DNA damage in cancer cells

To determine whether PRMTis influence DNA damage responses induced by PARPi treatments, a comet assay was used to measure DNA damage in a panel of HR-proficient cancer cells that had been treated with olaparib alone or in combination with PRMTis. Although the olaparib treatment alone induced DNA damage in all four HR-proficient cancer cell lines, the extent of the DNA damage increased significantly when cells were treated with a combination of PRMTis (GSK3368715 or GSK3235025) and olaparib (**Figure 3A, B** and Figure S3A). This observation was further confirmed by a western blot assay to detect the DNA double-strand break marker, the phosphorylation of H2AX (γ H2AX [S139]), among different treatment conditions (**Figure 3C**).

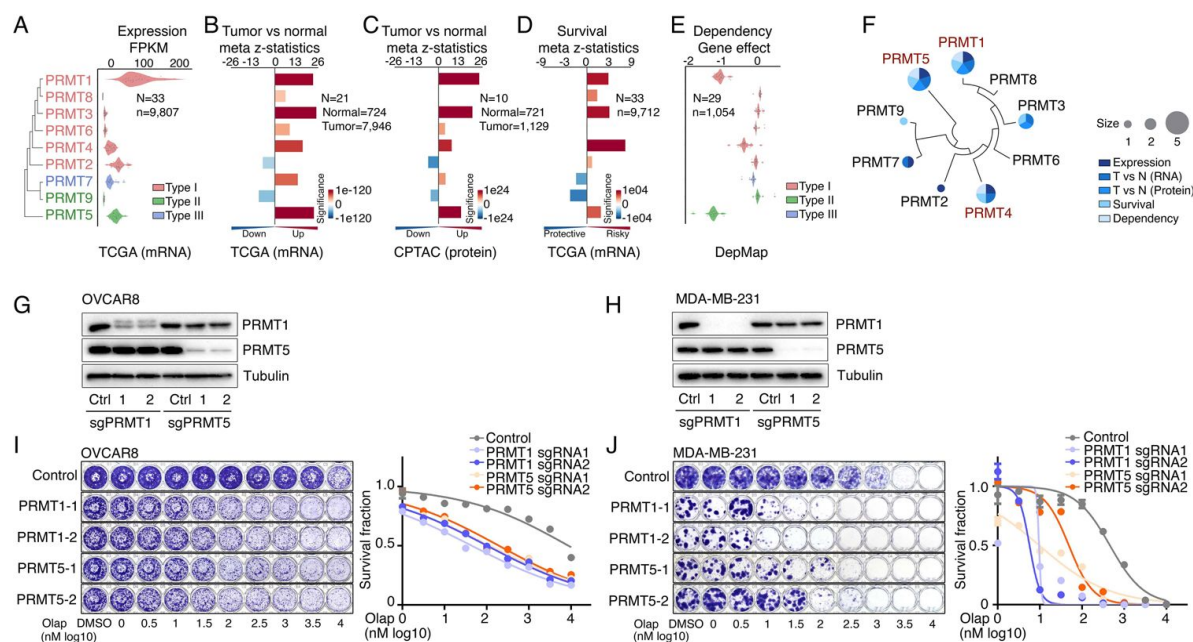


Figure 2.

PRMT1, PRMT4/CARM1, and PRMT5 are potential therapeutic targets in oncology.

A. mRNA expression levels of the PRMT family across the TCGA tumor samples. Each spot represents the average mRNA expression level of a cancer type ($n = 33$). **B.** and **C.** Meta z-statistics of mRNA (**B**) and protein (**C**) expression levels of the PRMT family in tumors compared with corresponding controls in the TCGA (**B**) or CPTAC (**C**) cohorts. Red and blue indicate up- and down-regulation, respectively. **D.** Meta z-statistics of the association between mRNA expression and patient outcome for the PRMT family across the TCGA cohort. Red and blue indicate poor and good outcomes, respectively. **E.** The dependency effects of the PRMT family based on the CRISPR screen from the DepMap project. Each spot represents the average gene effect of a cancer type ($n = 29$). A lower effect score indicates that a gene is more likely to be dependent in cancer cells. **F.** Summary of the above five features for each gene in the PRMT family. **G.** and **H.** Western blot analyses of PRMT1 and PRMT5 expression in OVCAR8 (**G**) and MDA-MB-231 (**H**) cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. **I.** and **J.** Sensitivity of olaparib in OVCAR8 (**I**) or MDA-MB-231 (**J**) cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Left panel: crystal violet staining of a colony formation assay; right panel: survival fraction quantified from the left panel.

Consistence with the enhanced DNA damage, the combination treatment significantly induced apoptosis, as measured by caspase-3/7 activity (**Figure 3D** [↗](#)), compared with the olaparib treatment alone. To exclude the potential off-target effects of PRMT chemical inhibitors, PRMT1 and PRMT5 were knocked out independently using CRISPR/Cas9 in OVCAR8 and MDA-MB-231 cells (**Figure 2G** [↗](#)). The specific knockout of PRMT1 or PRMT5 significantly enhanced the DNA damage induced by the PARPi treatment, thereby increasing apoptosis (**Figure 3E–3G** [↗](#) and **Figure S3B**). These observations suggested that the increased expressions of PRMT1 and PRMT5 in cancer may prevent cell death by enhancing DNA damage repair when tumor cells are exposed to endogenous or exogenous DNA damage-causing conditions. Consequently, we analyzed the correlations between the expression levels of PRMT1/5 and “50 hallmark” molecular signatures⁹³ [↗](#) in a large collection of cancer cell lines from the DepMap project (n = 1,200). Consistently, “DNA-repair” was a significant signature that positively correlated with both PRMT1 and PRMT5 expression levels in cancer cell lines (**Figure 3H** [↗](#)). Similar results were found in the primary tumor specimens across the TCGA cohort at individual cancer type and pan cancer levels (**Figure 3I** [↗](#), **3J** and **Table S6**). Supporting previous reports that the Myc-PRMT loop plays critical roles during tumorigenesis, “Myc-targets” signatures were the most significant signatures correlated with the expression of PRMT1/5 (**Figure 3H–J** [↗](#)).

PRMTs maintain expression of the genes associated with DDR and BRCAness

Given that PRMTs act as key epigenetic regulators through the modification of histone methylation, we investigated gene expression changes induced by PRMTi treatments. An RNA-seq analysis revealed that GSK3368715 and GSK3235025 treatments reduced protein-coding gene expression in MDA-MB-231 cells by approximately 12.7% and 6.9%, respectively (**Figure 4A** [↗](#)). Consistent with correlation analyses in cancer cell lines and primary tumors (**Figure 3H** [↗](#) to **3J** [↗](#)), a gene set enrichment analysis found that the downregulated genes in both the GSK3368715 and GSK3235025 treatment groups were significantly enriched in the DDR-associated pathways compared with controls (**Figure 4B** [↗](#)). The epigenetically repression of certain DDR genes induces the BRCAness phenotype, sensitizing cancer cells to PARPi treatments. Therefore, we collected the genes in the homologous recombination (HR) and Fanconi anemia (FA) pathways (two major DDR pathways affecting PARPi sensitivity)⁹⁴ [↗](#), ⁹⁵ [↗](#), as well as the genes that were previously reported to be associated with PARPi sensitivity⁹⁶ [↗](#). A total of 247 genes potentially implicated in PARP inhibitor sensitivity (i.e., BRCAness genes) were identified in a whole genome (**Figure 4C** [↗](#) and **Table S7**). As expected, many were downregulated by GSK3368715 (n = 29) and GSK3235025 (n = 31) treatments, with nine being significantly repressed by both of the PRMTis (**Figure 4A** [↗](#) and **Table S8**). This result was further validated by a qRT-PCR analysis (**Figure 4D** [↗](#)). Beyond epigenetic regulation, PRMTs play functional roles in RNA splicing through the methylation of RG/RGG motif-containing proteins in the splicing complexes. Consequently, we analyzed global RNA splicing changes between PRMTi treatments and controls. Consistent with previous reports, both the GSK3368715 and GSK3235025 treatments affected RNA splicing genome wide (**Figure 4E** [↗](#)). However, genes undergoing splicing changes were not significantly enriched in most DDR pathways (except for base excision repair in the GSK3235025 treatment) (**Figure 4F** [↗](#)), although splicing alterations were identified in certain BRCAness genes, such as ERCC1, after PRMTi treatments (**Figure 4G** [↗](#)). Unexpectedly, we found that a PARPi treatment alone also induced modest but significant splicing alterations compared with the control (**Figure 4E** [↗](#)). Combination treatments with GSK3235025, but not GSK3368715, significantly further enhanced splicing alterations compared with an olaparib or GSK3235025 treatment alone (**Figure 4E** [↗](#)). Thus, both PRMT1i and PRMT5i treatments repressed mRNA expression of a large number of BRCAness genes in cancer cells. Although some genes were inhibited by both inhibitors, PRMT1i and PRMT5i repressed distinct BRCAness genes. Splicing alterations in certain BRCAness genes were identified after a PRMTi treatment, and the PARPi-treatment-induced splicing susceptibility may also serve as a mechanism for the synergistic action of PRMTi and PARPi.

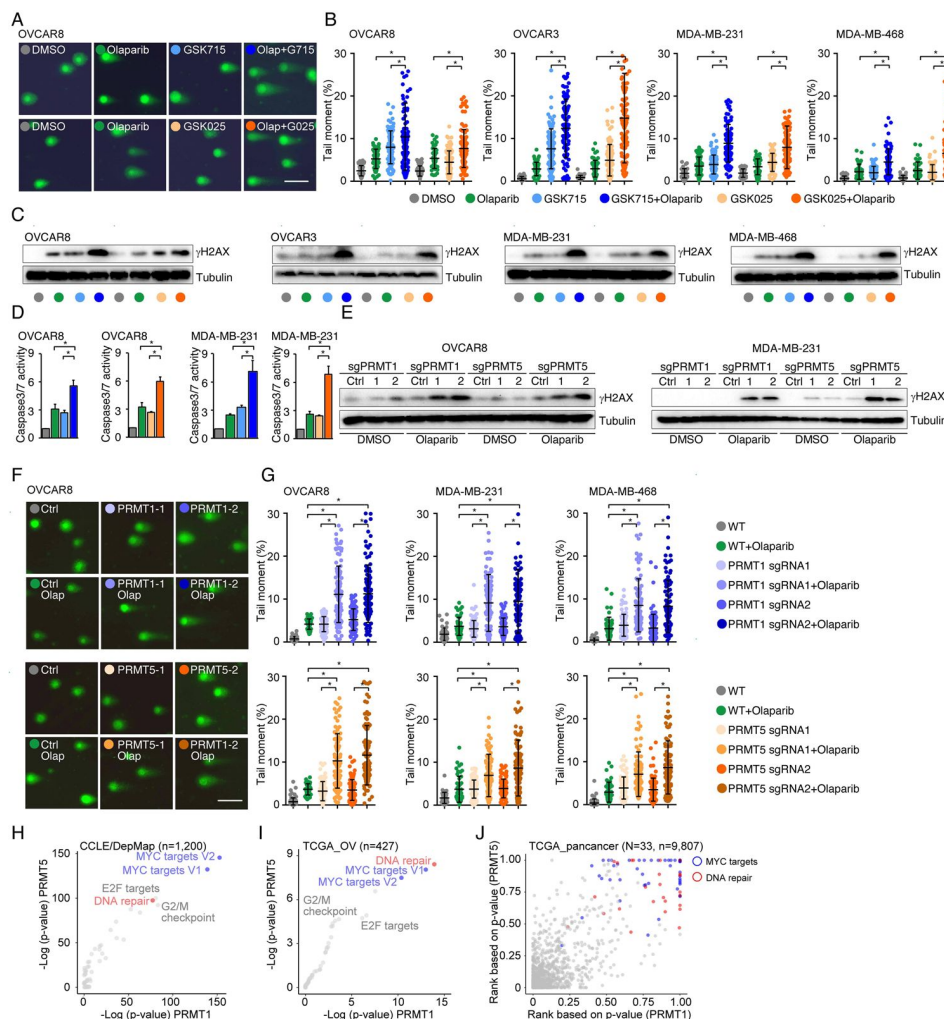


Figure 3.

PRMT inhibition enhances PARPi treatment-induced DNA damage in cancer cells.

A. A comet assay was used to measure DNA damage in cancer cells treated with DMSO, olaparib (5 μ M, 96 h), GSK3368715 (5 μ M, 96 h), GSK3235025 (5 μ M, 96 h), or combinations (representative images from OVCAR8). Scale bars, 10 μ m. **B.** The extent of DNA damage was quantified using the tail moment in comet assays of OVCAR8, OVCAR3, MDA-MB-231, and MDA-MB-468 cell lines treated with DMSO, olaparib, GSK3368715, GSK3235025, or combinations. Data are presented as means $\pm$ SDs, * p < 0.05 determined by two-tailed Student's t tests. **C.** Western blot analyses of γ H2AX in cancer cells treated with DMSO, olaparib, GSK3368715, GSK3235025, or combinations. **D.** Caspase-3/7 activity was measured using a caspase-Glo 3/7 assay in OVCAR8 and MDA-MB-231 cells treated with DMSO, olaparib, GSK3368715, GSK3235025, or combinations. Data are presented as means $\pm$ SDs, n = 3 biological replicates, * p < 0.05 determined by two-tailed Student's t tests. **E.** Western blot analyses of γ H2AX in OVCAR8 and MDA-MB-231 cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. **F.** A comet assay was used to measure olaparib treatment-induced DNA damage in cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9 (representative images from OVCAR8). Scale bars, 10 μ m. **G.** The extent of olaparib treatment-induced DNA damage was quantified using the tail moment in comet assays of OVCAR8, MDA-MB-231, and MDA-MB-468 cell lines in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Data are presented as means $\pm$ SDs, * p < 0.05 determined by two-tailed Student's t tests. **H.** and **I.** Correlations between the expression levels of PRMT1/PRMT5 and "50 hallmark" molecular signatures in the cancer cell lines from the DepMap (H) and primary tumor specimens from the TCGA (ovarian cancer was represented as an example, I) cohort. **J.** Correlations between the expression levels of PRMT1/PRMT5 and "50 hallmark" molecular signatures across primary tumor specimens from the TCGA cohort. For each cancer type, the correlations between PRMT1/5 expression and "50 hallmark" molecular signatures were ranked 0 to 1 based on the p -values, with 1 indicating the most significant signature. Red and blue circles indicate DNA repair and Myc signatures, respectively.

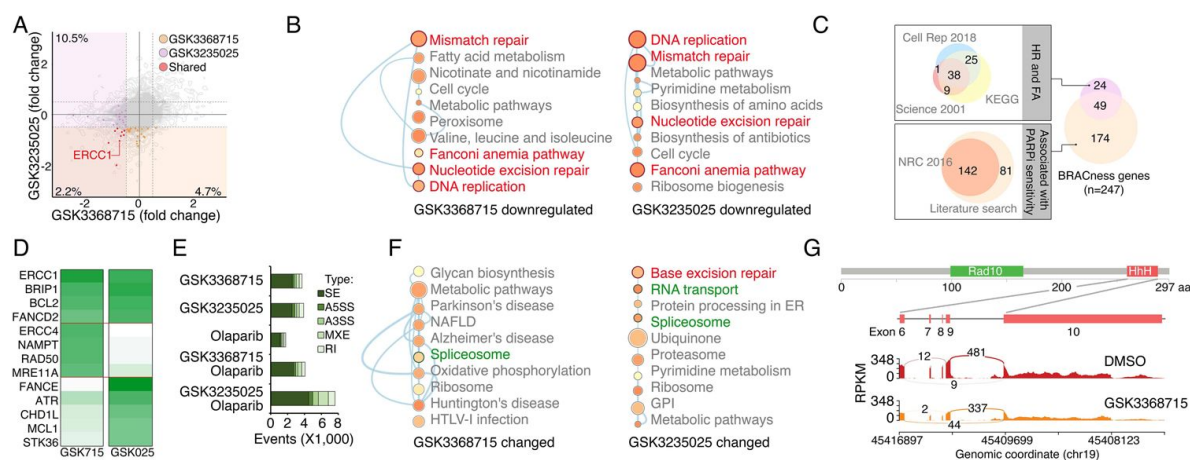


Figure 4.

PRMTis maintain the expression of genes associated with DNA damage repair (DDR) and BRCAness.

A. The dot plot shows the gene expression changes measured by RNA-seq in MDA-MB-231 cells. The BRCAness genes that were repressed by GSK3368715 and GSK3235025 treatments are highlighted with color. **B.** The top 10 KEGG pathways in which genes downregulated by GSK3368715 (left) and GSK3235025 (right) treatments were significantly enriched. Circle size indicates the fold enrichment. Lines indicate numbers of shared genes between pathways. The DDR pathways are highlighted in red. **C.** Illustration BRCAness gene collection method. First, the genes in two major DDR pathways associated with PARPi sensitivity (HR and FA) were collected from three independent databases. In total, 73 genes annotated by at least two databases were defined as HR/FA genes. Second, the genes associated with PARPi sensitivity were retrieved from a Nature Reviews paper (n = 142). An additional 81 genes were identified through a manual literature search using PubMed. In total, 223 genes were defined as PARPi sensitivity-associated genes. Taken together, 247 genes were defined as “BRCAness genes” in this study. **D.** The heatmap shows qRT-PCR validation of a group of BRCAness genes. Green color intensity indicates fold changes. **E.** The abnormal splicing event numbers induced by different treatments compared with the control treatment (DMSO) in MDA-MB-231 cells. Types of abnormal splicing events are indicated by different colors. **F.** The top 10 KEGG pathways in which abnormally spliced genes after GSK3368715 (left) and GSK3235025 (right) treatments were significantly enriched. Circle size indicates the fold enrichment. Lines indicate numbers of shared genes between two pathways. **G.** An abnormal splicing gene (ERCC1). Upper panel: the structure of ERCC1. The alternative splicing event was identified in exon 6-10, which encodes the helix-hairpin-helix (HhH) domain of ERCC1. Lower panel: Sashimi plots showing the exon truncation events of ERCC1 as detected by RNA-seq in MDA-MB-231 cells treated with DMSO and GSK3368715 independently. The X-axes indicate genomic locations, and the Y-axes indicate RNA-seq coverage. The numbers of junction reads are shown in the splice junctions (curved lines).

Repression of ERCC1 is a common mechanism by which PRMTs sensitize cells to PARPi treatment

To evaluate whether the BRCAness genes, which were repressed by PRMTs, play functional roles in synergistic action between PRMTs and PARPi, a customer-designed siRNA screen was performed in MDA-MB-231 cells (**Figure 5A**). An impact score for each candidate was estimated based on 1) expressional repression effects of PRMTs and 2) functional effects on PARPi sensitivity (**Figure 5B**). Among the candidates examined, ERCC1 had the highest impact score and was repressed by both GSK3368715 and GSK3235025 treatments (**Figure 5C**). In lung cancer, knocking out ERCC1 sensitizes cells to PARPi treatments⁹⁷, and its expression serves as a predictive biomarker for the platinum response⁹⁸. Therefore, we used ERCC1 as an example to evaluate whether transcriptional repression can serve as a mechanism. We found that, similar to PRMTs, ERCC1 expression was significantly upregulated across cancer samples compared with their corresponding control specimens (**Figure 5D**). Consistent with the association between PRMT1/5 and the DDR signature (**Figure 3**), significant positive correlations between PRMT1/5 and ERCC1 mRNA expression levels were observed in primary tumor specimens from the TCGA cohort (**Figure 5E**). These results suggest that elevated PRMT expression in cancer may serve as a mechanism to epigenetically maintain ERCC1 expression, which is critical to the survival of cancer cells that are subjected to high levels of endogenous DNA damage and increased replication-related stress. Using western blots, we demonstrated that PRMTi treatments repressed ERCC1 expression at the protein level in a dose-dependent manner (**Figure 5F**). As expected, under these treatment conditions, GSK3368715 and GSK3235025 reduced global aDMA and sDMA levels, respectively. Notably, in consistence with previous reports^{73, 99}, the GSK3368715 treatment also increased global sDMA levels (Figure S4), suggesting that PRMT1 and PRMT5 share a substrate (i.e., MMA), and/or their functions may be compensatory in certain cellular contexts^{73, 99}. These observations were also validated by genetically knocking out PRMT1 and PRMT5 independently using CRISPR/Cas9 (**Figure 5G** and Figure S4B). To determine whether the repression of ERCC1 by PRMTs was mediated by an epigenetic (transcriptional) mechanism, we cloned the ERCC1 promoter region (−1,500 to +120) and inserted it into a reporter vector (pGL3-basic). Both the chemical PRMTi treatment and genetic knockout of PRMT1/5 significantly reduced ERCC1 promoter activities in cells (**Figure 5H**). Next, we determined whether the PRMTi treatment was able to reduce the ERCC1 function. ERCC1 and XPF form a structure-specific nuclease that is required for the removal of ultraviolet (UV) radiation-generated cyclobutane pyrimidine dimers (CPDs). Thus, measuring the rate of UV-induced CPD removal serves as a semi-quantitative assay to monitor ERCC1 activity^{100, 101} (**Figure 5I**). The OVCAR8 cells were pretreated with GSK3368715 (2.5 μ M) and GSK3235025 (5 μ M) independently for 48 h, then CPDs were induced by exposure to UV radiation (15 J/m², **Figure 5I**). Using a specific CPD antibody (Kamiya Biomedical Company, MC-062)^{100, 101}, immunofluorescent staining demonstrated that PRMTi treatments significantly decreased the CPD removal rates compared with controls after UV exposure (**Figure 5J**). This observation was validated by genetically knocking out PRMT1 or PRMT5 using CRISPR/Cas9 (**Figure 5K**). Additionally, a DNA-blotting assay confirmed the immunofluorescent staining results (**Figure 5L, M**). To further demonstrate the functional involvement of ERCC1 in the synergistic actions between PRMTs and PARPi, we knocked out ERCC1 in cancer cell lines using CRISPR/Cas9 (**Figure 5N**). As expected, the ERCC1-XPF activity was significantly reduced (Figure S4C). Colony formation assays demonstrated that ERCC1 knockout cancer cell lines were sensitized to PARPi treatments (**Figure 5O**). Finally, the forced ERCC1 expression by lentiviral infection partially reduced the synergistic actions between PRMTs and PARPi in these cell lines (**Figure 5P**). Therefore, both type I PRMT and PRMT5 inhibitor treatments significantly repressed ERCC1 expression and activity levels in cancer cells. ERCC1 serves as a common mechanism by which PRMTs sensitize cells to PARPi treatments.

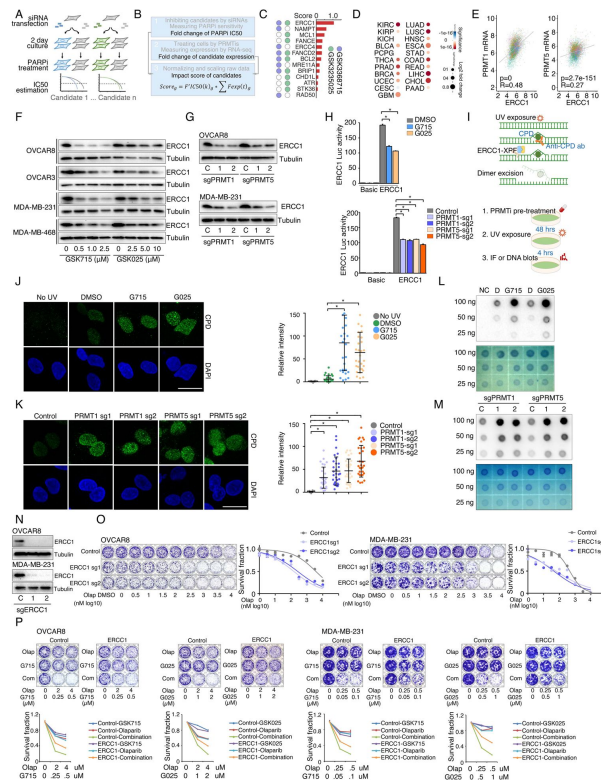


Figure 5.

ERCC1 repression is a common mechanism by which PRMTs sensitize cells to PARPi treatment.

A. A siRNA screen was performed to evaluate the effects of knocking down the candidate genes on PARPi sensitivity in MDA-MB-231 cells. Gray: control siRNA transfection. **B.** Schematic of the impact score estimation for each candidate gene based on RNA-seq and a siRNA screen. **C.** Summary of the impact scores of 13 examined candidates. **D.** Expressional differences in ERCC1 mRNA levels between tumor specimens and their corresponding controls. Red and blue indicate up- and down-regulation, respectively. Size: fold change, intensity: p-value. **E.** Correlations of mRNA expression levels between ERCC1 and PRMT1 (left) and PRMT5 (right) across all the TCGA tumors. Color indicates cancer type. **F.** Western blot analysis of ERCC1 protein expression in cancer cells treated with different GSK3368715 and GSK3235025 concentrations, independently, for 72 h. **G.** Western blot analysis of ERCC1 protein expression in OVCAR8 and MDA-MB-231 cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. **H.** Luciferase reporter assay of ERCC1 core promoter activity in HEK293T cells treated with GSK3368715 (5 μ M, 48 h) and GSK3235025 (5 μ M, 48 h) independently (upper panel), or after knocking out PRMT1 and PRMT5, independently (lower panel). Statistical difference was assessed using Student's t test, * $p < 0.05$; $n = 3$. Error bars represent means $\pm$ SDs. **I.** Schematic of UV-induced cyclobutane pyrimidine dimers (CPD) detection by antibody (upper panel) and the experimental schedule (lower panel). **J.** Representative images (left) and quantitative results (right) of UV-induced CPD formation in MDA-MB-231 cells treated independently with DMSO, GSK3368715 (2.5 μ M, 96 h), and GSK3235025 (5 μ M, 96 h). Cells not subjected to UV treatment were used as the negative control. Scale bars, 10 μ m. Statistical analysis by Student's t test, * $p < 0.05$; $n = 3$. Error bars represent means $\pm$ SD. **K.** Representative images (left) and quantitative results (right) of UV-induced CPD formation in MDA-MB-231 cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Scale bars, 10 μ m. Statistical differences were assessed using Student's t test, * $p < 0.05$; $n = 3$. Error bars represent means $\pm$ SDs. **L.** Dot blots of CPD levels in MDA-MB-231 cells treated independently with DMSO, GSK3368715 (5 μ M, 96 h), and GSK3235025 (5 μ M, 96 h). Cells not subjected to UV treatment were used as the negative control. Methylene blue staining was used as the loading control. **M.** Dot blots of CPD levels in MDA-MB-231 cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Methylene blue staining was used as the loading control. **N.** Western blot analysis of ERCC1 protein expression in OVCAR8 and MDA-MB-231 cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. **O.** Sensitivity of OVCAR8 and MDA-MB-231 cells to olaparib. Left panel: crystal violet staining of a colony formation assay; right panel: survival fraction quantified from the upper panel. **P.** Sensitivity of OVCAR8 and MDA-MB-231 cells, which overexpressed ERCC1 owing to a lentiviral infection, to olaparib, GSK3368715, GSK3235025, and combinations. Upper panel: crystal violet staining of a colony formation assay; lower panel: the survival fraction quantified from the upper panel.

PRMT inhibition activates intrinsic innate immune pathways in cancer cells

We also analyzed the PRMTi treatment-induced genes using a gene set enrichment analysis and found that the most upregulated pathways were associated with immune responses (Figure S5A). Using the immune-related gene list from an immune gene annotation database (ImmPort), 516 reliably detectable genes (RPKM >1 by RNA-seq) in MDA-MB-231 cells were defined as the immune-related genes. Consistently, we found that 17.8% and 23.6% were upregulated by GSK3368715 and GSK3235025 treatments, respectively (Figure S5B). Compared with the upregulated non-immune-related genes, a significant number of the immune-related genes were induced by PRMT inhibition (GSK3368715, $p = 7.6 \times 10^{-9}$; GSK3235025, $p = 1.8 \times 10^{-18}$). Notably, 53 immune-related genes were upregulated by both PRMTis, and 37.7% of them were classified as innate immune-related genes by the InnateBD database (Figure S5C). These innate immune-related genes were associated the major innate immune signaling pathways, such as cytokines, chemokines, and their receptors (Figure S5D). These results suggest that PRMTi treatments may enhance intrinsic immune reactions in tumor cells, providing a strong rationale for the application of PRMTis (or in combination with immune checkpoint inhibitors) in immuno-oncology. Notably, PARPi treatments trigger anticancer innate immune responses by increasing cytosolic DNA. Consequently, using the phosphorylation of TBK1 as a functional marker to monitor the activities of innate immune signals in tumor cells, we found that the PRMTi treatment alone induced comparable levels of p-TBK1 as the PARPi treatment, whereas the combination of PRMTi and PARPi increased p-TBK1 remarkably compared with single agent treatments (Figure S5E). This observation was confirmed by genetically knocking down PRMT1 and PRMT5 independently using CRISPR/Cas9 (Figure S5F). Because both PRMTi and PARPi treatments induced DNA damage in cancer cells, we also examined cytoplasmic micronuclei using PicoGreen staining, which specifically recognizes dsDNA (Figure S5G). Indeed, the numbers of cytoplasmic micronuclei significantly accumulated when cells were treated with PRMTis or PARPi alone, whereas the combination treatment synergistically enhanced this response (Figure S5H). The combination of PRMTis and PARPi may synergistically enhance anticancer immunity because cytoplasmic dsDNA triggers the cGAS-STING-TBK1 pathway¹⁰².

Type I PRMT and PRMT5 inhibition act synergistically to enhance PARPi sensitivity

The inhibition of type I PRMT and PRMT5 represses shared, as well as distinct, BRCAness genes (Figure 4A); consequently, we hypothesized that the combination of type I PRMT and PRMT5 inhibitors may act synergistically to enhance PARPi sensitivity. Thus, using ERCC1 as an example, we examined whether there was a synergistic effect on the repression of shared targets. Both chemical (PRMTis) and genetic (CRISPR) approaches demonstrated that the repression of both PRMT1 and PRMT5 more significantly reduced ECRR1 promoter activities than single treatments (Figure 6A, B). This transcriptional synergistic action was further confirmed by qRT-PCR (Figure 6C). More importantly, the combination treatment almost completely repressed ERCC1 protein expression in cancer cell lines (Figure 6D). Using combinations of inhibitors that target the same pathway or biological process not only leads to synergistic effects and overcomes resistance, it may also reduce the required dose of each drug, thereby reducing potential side effects and increasing the therapeutic window. For example, co-targeting type I PRMTs and PRMT5 using a combination of different PRMTis showed synergistic actions and promising therapeutic efficacies in preclinical models^{48, 73, 74}. Because PARPis are well tolerated in clinical practice, we next evaluated whether de-escalation doses of PRMTis in combination with PARPi can achieve a therapeutic efficacy in a panel of cancer cells (Figure 6E). Decreased independent doses of GSK3368715 (125 nM) and GSK3235025 (500 nM) achieved identical synergistic effects as the original doses of these two inhibitors (Figure 6E). Furthermore, although extremely low doses of GSK3368715 (25 nM) and GSK3235025 (100 nM) independently showed moderate synergy with

olaparib, and a triple combination produced a strong synergistic action with olaparib. More importantly, when we further reduced doses of PRMTi, synergistic effects were still achieved (i.e., GSK3368715 at 12.5 nM and GSK3235025 at 12.5 nM) in the triple combination setting (**Figure 6F**). Similar results were observed for another FDA-approved PARPi, rucaparib (**Figure 6G** and Figure S6). Finally, we validated our *in vitro* findings through *in vivo* experiments using immunodeficient nude mouse models treated with low doses of PRMT inhibitors (**Figure 6H–K**). Consistently, olaparib in combination with GSK3368715 and GSK3235025 independently significantly reduced tumor growth *in vivo* compared with the olaparib treatment alone, although in this low-dose PRMTi setting, PRMTi alone showed no (GSK3235025) or modest (GSK3368715) effects on tumor growth. In addition, the triple combination achieved a stronger therapeutic effect compared with the double combinations. Notably, consistent with previous report⁷³, combinations of GSK3368715 and GSK3235025 also showed promising therapeutic effects compared with GSK3368715 and GSK3235025 alone. Importantly, no significant side effects were observed in PRMTi and PARPi combinations and the triple combination in this low-dose PRMTi setting (**Figure 6L**), suggesting that both double and triple combinations may be tolerable in future clinical developments.

Discussion

An “induced PARPi sensitivity by epigenetic modulation” strategy (i.e., epigenetically impairing the genes of the DDR pathway in HR-proficient tumors to induce an HR-deficiency or other DDR-defects, thereby enhancing and sensitizing tumor cells to PARPi) has been developed recently to expand PARPi clinical applications^{9, 10, 15}. For example, our previous study has demonstrated that repression of BET activity can significantly reduce the expression levels of essential HR genes by inhibiting the transcriptional activities of their super enhancers¹⁵. This preclinical discovery is being evaluated in a Phase 1b/2 clinical trial (NCT03901469) by a combination of BETi (ZEN-3694) with PARPi (talazoparib) for treatment of patients with metastatic TNBC without germline BRCA1/2 mutations (i.e., intrinsic HR-proficient patients)³⁴. Excitingly, promising anti-cancer activity of this novel drug combination has been observed in the on-going clinical trial. Importantly, paired tumor biopsies have shown robust target engagement (repression of BRCA1 and RAD51 mRNA expression)³⁴, consistent with our preclinical studies in xenograft tumor models¹⁵. This trial is being expanded to a Phase 2b stage to accrue additional patients with TNBC. In addition, other combination approaches^{15–33} based on the “induced PARPi sensitivity by epigenetic modulation” strategy also achieved promising clinical responses in intrinsic or acquired HR-proficient settings^{23, 28–31, 34}. Most importantly, due to cancer cells being extremely sensitive to the epigenetic repression of DDR genes compared with normal cells, the clinical studies demonstrated that this strategy is well tolerable for the patients with cancer.

However, two key clinically challenges still need to be addressed for future expansion of the application of this strategy in oncology: i.e., which clinically applicable epi-drug(s) can provide the strongest synergistic effects with PARPi? and can a BRCA-independent strategy be developed? To design and evaluate highly effective and BRCA-independent epi-drug/PARPi combination strategy, we performed a drug screen by combining olaparib, an FDA-approved PARPi, with a set of clinical applicable epi-drugs, which contain 74 well-characterized epigenetic modulators that target five major classes of epigenetic enzymatic actions. Among them, 7 are FDA-approved drugs in oncology, and 14 and 54 are in clinical trials and preclinical development, respectively. Notably, both type I PRMTi and PRMT5i exhibited high combination and clinical priority scores in our screen. This result was further confirmed by using different FDA-approved PARPis in multiple HGSOC and TNBC cell lines. Our functional studies demonstrated that PRMT inhibition significantly enhanced PARPi treatment-induced DNA damage in HR-proficient cancer cell lines. In

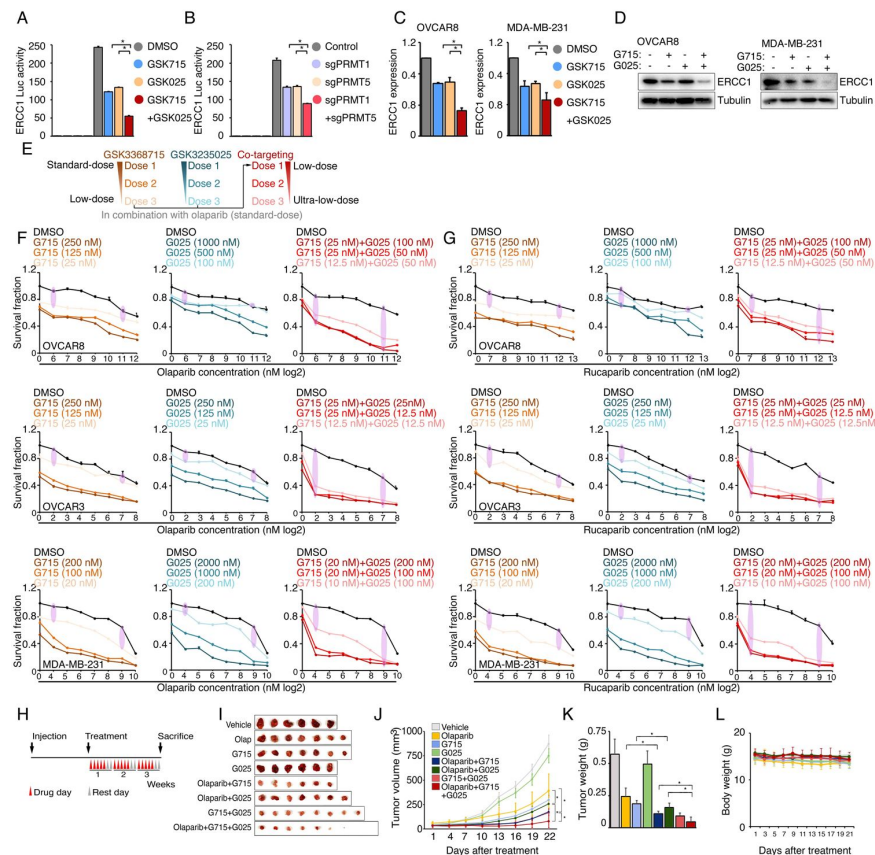


Figure 6.

Type I PRMT and PRMT5 inhibition acts synergistically to enhance PARPi sensitivity.

A. Luciferase reporter assay of the ERCC1 core promoter activity levels in HEK293T cells treated with DMSO, GSK3368715 (5 μ M, 48 h), and GSK3235025 (5 μ M, 48 h) independently, or with the combination of GSK3368715 (5 μ M, 48 h) and GSK3235025 (5 μ M, 48 h). Statistical differences were assessed using Student's *t* test, **p* < 0.05; *n* = 3. Error bars represent means $\pm$ SDs. **B.** Luciferase reporter assay of the ERCC1 core promoter activity levels in HEK293T cells in which PRMT1, PRMT5, and PRMT1/5 were independently knocked out using lentiviral CRISPR/Cas9. Statistical differences were assessed using Student's *t* test, **p* < 0.05; *n* = 3. Error bars represent means $\pm$ SDs. **C.** qRT-PCR analysis of ERCC1 RNA expression in OVCAR8 and MDA-MB-231 cells treated with GSK3368715 (5 μ M) and GSK3235025 (5 μ M) independently for 48 h. **D.** Western blot analysis of ERCC1 protein expression in OVCAR8 and MDA-MB-231 cells treated with GSK3368715 (5 μ M) and GSK3235025 (5 μ M) independently for 72 h. **E.** Dose de-escalation scheme. The lowest doses of GSK3368715 and GSK3235025 independently for PRMTi and the PARPi combination were first identified. They were then used as initial doses for triple combination. The doses of GSK3368715 and GSK3235025 were further reduced until the lowest doses for each PRMTi in the triple combination was identified. **F.** Sensitivity levels of OVCAR8 (upper panel), OVCAR3 (middle panel), and MDA-MB-231 (lower panel) cells to olaparib alone or olaparib combined with three different doses of GSK3368715 (left panel), GSK3235025 (middle panel), and GSK3368715 plus GSK3235025 (right panel). **G.** Sensitivity levels of OVCAR8 (upper panel), OVCAR3 (middle panel), and MDA-MB-231 (lower panel) cells to rucaparib alone or rucaparib combined with three different doses of GSK3368715 (left panel), GSK3235025 (middle panel), and GSK3368715 plus GSK3235025 (right panel). **H.** Schematic of the MDA-MB-231 xenograft experimental design. MDA-MB-231 cells were implanted subcutaneously in immunodeficient nude mice and grown until tumors reached a size of approximately 30 mm³. Tumor-bearing mice were randomized and then received vehicle, 50 mg/kg olaparib i.p., 25 mg/kg GSK3368715 p.o., 50 mg/kg GSK3235025 p.o., or combinations of two agents or three agents, 5 days a week for 3 weeks. Caliper measurements were taken every other day from the initiation of drug treatment. **I.** Images of tumors collected from animals receiving vehicle, olaparib, GSK3368715, GSK3235025, and combinations of two agents or three agents. **J.** Tumor growth curve for each treatment group in immunodeficient nude mice. Statistical significance was assessed using a two-tailed Student's *t* test. Error bars represent means $\pm$ SDs. **K.** Bar plot showing the mean tumor volume of each treatment group. Statistical significance was assessed using a two-tailed Student's *t* test. Error bars represent means $\pm$ SDs. **L.** Changes in the body weights of different murine treatment groups. Statistical significance was assessed using a two-tailed Student's *t* test. Error bars represent means $\pm$ SDs.

consistency with the significant and positive correlation between the expression of PRMTs and DDR genes across primary cancer specimens, we observed that PRMT activities maintain the transcription of genes associated with DNA damage repair and “BRCAness” in cancer cells.

The expression levels of many PRMT family members are significantly elevated in cancer cells and associated with poor clinical outcomes, strongly suggesting that they may serve as a class of novel therapeutic targets in oncology. Our large-scale genomic and functional profiles from TCGA and DepMap further confirm that PRMT1, PRMT4, and PRMT5 are potential therapeutic targets in oncology. Major pharmaceutical companies have developed more than ten selective and potent PRMTis that are currently undergoing early-stage clinical evaluation. Development of novel treatment strategy to expand its potential clinical application is a key challenge for application of PRMTis in oncology. Given that recent clinical trials indicate that the activity of PRMTi alone may be insufficient to manage patients with cancer^{81–88}, even in MTAP-deletion setting, therefore, the design and evaluation of combination approaches using PRMTi with other therapeutic drugs are urgently needed. Our preclinical studies provide a strong rationale for a combination of PRMTi with PARPi for treatment of patients with HR-proficient HGSOC or TNBC. In addition, this strategy may also be expanded to other cancer types such as prostate and pancreatic cancers, in which PARPi has been approved by the FDA for clinical management.

As major histone modifiers, PRMTs control methylation of histones, thereby epigenetically regulating gene transcription and expression^{35–39}. Additionally, through protein post-translational modification, PRMTs also directly regulate non-histone protein functions. For examples, PRMTs preferentially modify the splicing factors to maintain appropriate splicing process in cells. Given that the PRMT-regulome broadly influences multiple cellular processes in cancer cells, the mechanisms of action for the PRMT and PARP inhibitor combination may be highly complicated and context-dependent. For example, it has been reported that PRMTs indirectly regulate DDR gene expression through their functions on maintaining arginine methylation of the proteins in splicing complex^{59, 62, 66}. Consistent with these studies, we also found that splicing of certain DDR genes, such as ERCC1, was impaired by PRMT inhibition. Additionally, PRMTs may also directly impact arginine methylation in certain DDR genes, thereby influencing their function in DNA damage response^{52–58, 60, 61, 63, 65}. In the context of HGSOC/TNBC, our results suggest that treatment with both type 1 PRMT and PRMT5 inhibitors suppressed RNA expression of many genes potentially involved in PARP inhibitor sensitivity (i.e. BRCAness genes). Although some of them were inhibited by both inhibitors, type 1 PRMT and PRMT5 inhibitors repressed distinct BRCAness genes. Splicing alterations in certain BRCAness genes were identified after a PRMTi treatment, therefore, the PRMTi-treatment-induced splicing susceptibility may also serve as an alternative mechanism for the synergistic action of PRMTi and PARPi. However, BRCAness genes were not significantly enriched among the genes whose splicing was significantly altered by PRMTi treatment, suggesting that splicing alteration might not serve as a dominant mechanism for PRMTi/PARPi combination, at least in the HGSOC/TNBC setting. Our studies not only provided mechanistic explanations for PRMTi and PARPi combinations but also provided novel information for the clinical development of this strategy. For example, ERCC1 expression may serve as a predictive biomarker for patient selection. Notably, ERCC1 is being evaluated in the clinic as a novel biomarker for predicting platinum sensitivity in lung cancer^{98, 103–105}. Finally, tumor intrinsic genomic status may affect responses to PRMTi alone or combination treatments^{35–39}. Supporting previous reports found that the Myc-PRMT loop plays critical roles during tumorigenesis⁴⁵, and “Myc-targets” signatures were the most significant signatures correlated with the expression of PRMT1 and PRMT5. Therefore, under MYC-hyperactivated condition, tumors may be extremely sensitive to PRMTis or PRMTi/PARPi combination.

Co-targeting type I PRMT and PRMT5 has shown strong synergistic actions and promising therapeutic efficacies^{48, 73, 74}. Combinations of drugs that target the same pathway (e.g., combination of dabrafenib and trametinib) not only lead to synergistic effects and overcome

resistance, they may also reduce the required dose of each drug, decreasing side effects and increasing the therapeutic window⁹⁹. In a PARPi treatment setting, type I PRMT and PRMT5 regulated shared as well as distinct BRCAness genes, suggesting that co-targeting type I PRMT and PRMT5 may produce a synergistic effect. Indeed, we observed that dual treatments of type I PRMT and PRMT5 more significantly reduced ECRR1 expression compared with single treatments. Notably, in consistence with previous reports^{73, 99}, PRMT1i increased global sDMA levels, which suggested that type I PRMT and PRMT5 share a substrate (i.e., MMA) and/or their functions are compensatory. Therefore, we believe that co-targeting type I PRMT and PRMT5 not only results in a synergistic effect that enhances PARPi sensitivity, reducing the required dose of each drug, but it also serves as a strategy to overcome potential resistance due to the compensatory functions of type I PRMT and PRMT5. This concept was supported by our *in vitro* triple combination experiments as well as *in vivo* studies. Although treatment-related adverse events are manageable and reversible with treatment interruption and reduction, efficacy of PRMTis as monotherapeutic agents was modest in clinical trials^{81–88}. Therefore, novel combination strategies using lower doses of PRMTis urgently need to be designed and evaluated. Our studies indicated that, because PRMT “addiction” enables tumor cells to survive DNA damage, targeting PRMTs can lead to “treatment-induced DNA damage sensitivity”, thereby sensitizing tumor cells to PARPi (and other DNA damaging drugs) treatments. A low-dose PRMTi in combination with PARPi may represent a more tolerable strategy for future clinical applications in oncology. Finally, co-targeting type I PRMT and PRMT5 may further significantly reduce the doses of each PRMTi (i.e., ultra-low-dose), thereby decreasing side toxicity effects of PRMTis. Taken together, our studies provide a strong rationale for clinical application of a combination of PRMT and PARP inhibitors in the patients with HR-proficient ovarian or breast cancer. Our mechanistical characterization may contribute to the future clinical development of PRMTis by defining mechanism of action, identifying rationales for combination, selecting response predictive biomarkers, and guiding dose and treatment schedules.

In summary, our study identifies PRMT1 and PRMT5 as key epigenetic regulators of DNA damage repair and shows that their inhibition sensitizes HR-proficient tumors to PARP inhibitors by repressing transcription and altering splicing of BRCAness genes. Distinct from prior strategies, dual inhibition of type I PRMT and PRMT5 exhibits strong synergy, allowing for lower-dose combination treatments that may reduce toxicity. Our findings also nominate ERCC1 as a potential predictive biomarker and suggest that MYC-driven tumors may be particularly responsive to this approach. Collectively, these results offer a mechanistic rationale and translational framework to broaden the clinical application of PARP inhibitors.

Materials and Methods

Cell culture

OVCAR8, OVCAR3, MDA-MB-231 and MDA-MB-468 cells were purchased from the ATCC or the NCI Development Therapeutics Program without further authentication. All cell lines were maintained at 37°C and 5% CO₂. Cells were cultured in RPMI1640 medium (Invitrogen) supplemented with 10% fetal bovine serum (FBS, Invitrogen). Cells were routinely tested for mycoplasma contamination using Mycoplasma Plus PCR Primer Set (Agilent, Santa Clara, CA).

Drug combination screening and validation

OVCAR8 and MDA-MB-231 cells were used for drug combination screening. Before screening, each epigenetic inhibitor (n=74) was arrayed in 48-well plates at concentrations of 0.1 μM, 1 μM, 10 μM for IC₅₀ calculation. For screening, cells were seeded in 48-well plates at a concentration of 3,750 cells/mL for OVCAR8 and 625 cells/mL for MDA-MB-231 in 400 μL of medium per well, and then treated with a single drug or with a combination of olaparib and one of the 74 epigenetic drugs 10 days for OVCAR8 or 8 days for MDA-MB-231. When testing for synergy of olaparib with epigenetic

drugs, the drug combinations were plated with 5-point serially constant two-fold ratio dilutions in 48-well plates using Eppendorf Xplorer plus Electronic Single Channel Pipette. Drugs were changed every two days. Cell viability testing was performed using the crystal violet staining assay (Sigma). After crystal violet was dissolved in 2% SDS, absorbance value was measured at 590 nm with spectrophotometer. Combination index (CI) and fraction affected (Fa) values were calculated using Compusyn software¹⁰⁶. The synergistic action between PRMTis and PARPis were further validated in a panel of cancer cell lines. In addition, triple drug combination was also evaluated in the same cell lines. Detailed combination condition was listed in Table S9.

Colony formation assay

Cells were seeded in 48-well plates at a concentration of 3,750 cells/mL for OVCAR8, 10,000 cells/mL for OVCAR3, 625 cells/mL for MDA-MB-231 and 10,000 cells/mL for MDA-MB-468 in 400 μ L of medium per well overnight. Cells were then treated with a single drug or a drug combination. After fixation with methanol, colonies were stained with 0.5% crystal violet for 30 min. For quantification, crystal violet was dissolved in 2% SDS and read by microplate reader.

CRISPR/Cas9 knockout

LentiCRISPRv2 was purchased from Addgene (plasmid #98290). sgRNAs targeting PRMT1, PRMT5 and ERCC1 were ordered from IDT and cloned to LentiCRISPRv2. LentiCRISPRv2 and packaging vectors were transfected into 293T cells. The medium was changed 8 hr after transfection, and the medium containing lentivirus was collected 48 hr later. Cancer cells were infected with lentivirus in the presence of 8 μ g/ml polybrene. 24 hours post-infection, infected cells were selected with 1 μ g/mL puromycin and the surviving pools were expanded for the following experiments.

Protein isolation and Western blot

Western blotting was performed using the following primary antibodies: anti-PRMT1 (Cat No: 2449S, CST); anti-PRMT5 (Cat No: 79998S, CST); anti-aDMA (Cat No: 13522S, CST); anti-sDMA (Cat No: 13222S, CST); anti-ERCC1 (Cat No: 3885S, CST); anti-phospho-H2AX (S139) (Cat No: 05-636, Clone No: JBW301, Millipore); p-TBK1 (Cat No: 5483S, CST) and anti- β -Tubulin (Cat No: 2128, Clone No: 9F3, CST), followed by incubation with secondary antibodies conjugated with horseradish peroxidase (HRP, GE Healthcare Life Sciences). Immunoreactive proteins were visualized using the LumiGLO chemiluminescent substrate (Cell Signaling).

Comet assays

For PRMT1/5 inhibitors treatment comet assay, OVCAR8 cells were treated with DMSO, 5 μ M olaparib, 5 μ M GSK3368715, 5 μ M olaparib combined with 5 μ M GSK3368715, 5 μ M GSK3235025 and 5 μ M olaparib combined with 5 μ M GSK3368715 for 96 hours. OVCAR3 cells were treated with DMSO, 1 μ M olaparib, 1 μ M GSK3368715, 1 μ M olaparib combined with 1 μ M GSK3368715, 1 μ M GSK3235025 and 1 μ M olaparib combined with 1 μ M GSK3368715 for 96 hours. MDA-MB-231 cells were treated with DMSO, 5 μ M olaparib, 2.5 μ M GSK3368715, 5 μ M olaparib combined with 2.5 μ M GSK3368715, 5 μ M GSK3235025 and 5 μ M olaparib combined with 5 μ M GSK3368715 for 96 hours. MDA-MB-468 cells were treated with DMSO, 2.5 μ M olaparib, 2.5 μ M GSK3368715, 2.5 μ M olaparib combined with 2.5 μ M GSK3368715, 5 μ M GSK3235025 and 2.5 μ M olaparib combined with 5 μ M GSK3368715 for 96 hours. For PRMT1/5 KO comet assay, wide type, PRMT1 sgRNA KO and PRMT5 sgRNA KO cells were treated with 5 μ M olaparib for 96 hours. Neutral comet assays with SYBR gold staining (Invitrogen) were performed. The quantification of tail DNA was done using CASP software.

Immunofluorescent staining

For CPD foci staining, DMSO, 2.5 μ M GSK3368715 (MDA-MB-231) or 5 μ M GSK3368715 (OVCAR8), 5 μ M GSK3235025 treated cells or PRMT1/5 KO cells were irradiated with UV (15 J/m²) and then allowed to repair for indicated times in fresh medium. Cells were fixed in solution containing 3% paraformaldehyde and 2% sucrose for 10 min at room temperature. Cells were subsequently permeabilized with 0.5% Triton solution for 5 min at 4°C and then incubated with anti-CPD antibody (Cat No: CAC-NM-DND-001, Cosmo Bio LTD) in PBST buffer (PBS plus 0.1% Tween-20, 0.02% NaN₃) overnight at 4°C. Cells were then washed three times with PBST and then incubated with goat anti-mouse IgG cross-adsorbed secondary antibody, Alexa Fluor 488 (Cat No: A-11017, Thermo) for 1 hour at room temperature. After four washes with PBST, coverslips were mounted onto glass slides using Vectashield mounting medium containing DAPI (Vector Laboratories) and visualized using an Axiovert 200M inverted microscope (Zeiss). For PicoGreen micronuclei staining, DMSO, 2.5 μ M GSK3368715 (MDA-MB-231) or 5 μ M GSK3368715 (OVCAR8), 5 μ M GSK3235025 treated cells were fixed in solution containing 3% paraformaldehyde and 2% sucrose for 10 min at room temperature. Cells were subsequently permeabilized with 0.5% Triton solution for 5 min at 4°C and then incubated with PicoGreen (1:1000 dilution) (Cat No: P11496, ThermoFisher Scientific) in PBST buffer (PBS plus 0.1% Tween-20, 0.02% NaN₃) overnight at 4°C. After four washes with PBST, coverslips were mounted onto glass slides using Vectashield mounting medium containing DAPI (Vector Laboratories) and visualized using an Axiovert 200M inverted microscope (Zeiss).

RNA isolation and qRT-PCR

Total RNA was extracted using TRIzol Reagent (Invitrogen) and reverse transcribed using the High Capacity RNA-to-cDNA Kit (Applied Biosystems). cDNA was quantified by an ABI ViiA 7 System (Applied Biosystems). QPCR primers were listed in Table S10.

RNA-sequencing

MDA-MB-231 cells were treated with DMSO, olaparib (5 μ M, 6 days), GSK3368715 (2.5 μ M, 6 days), GSK3235025 (5 μ M, 6 days) or combinations. Following total RNA extraction, libraries were prepared using TruSeq Stranded mRNA Library Prep Kit (Illumina), and equimolar libraries were multiplexed and sequenced on an Illumina NextSeq 500 (pair end 100 bp reads) by the BGI Genomics. FASTQ files were downloaded for the analysis.

Splicing analysis

Paired end RNA-seq profiles were used to characterize treatment-induced splicing changes. Differential splicing was detected by rMATS algorithm (<http://rnaseq-mats.sourceforge.net/download.html>), using a hierarchical framework to model exon inclusion levels¹⁰⁷,¹⁰⁸. Five alternative splicing events were identified, including skipped exon (SE), alternative 5' splice site (A5SS), alternative 3' splice site (A3SS), mutually exclusive exons (MXE), and retained intron (RI).

ERCC1 overexpression

For ERCC1 overexpression, OVCAR8 and MDA-MB-231 cells were seeded in 12-well plates at a confluence of 50% one day before transduction. Cells were transduced with lentiviral CD513B vector (System Biosciences) or CD513B-ERCC1. Medium was changed the next day. Cells were selected with 5 μ g/ml puromycin for 3 days, then seeded for sensitivity detection of olaparib alone, GSK3368715 alone/GSK3235025 alone, or combination of olaparib with GSK3368715 or GSK3235025.

DNA dot-blot

Cells were irradiated with UV (15 J/m²) and then allowed to repair for indicated times in fresh medium. Genomic DNA was extracted from these UV-irradiated cells using the QIAamp DNA Blood Mini Kit (51104, Qiagen) and RNase A (19101, Qiagen). DNA solutions were prepared in 50 μ L of PBS per sample to have 100 ng, 50 ng and 25 ng of genomic DNA. DNA solutions were spotted onto Hybond-N+ positively charged nylon membrane (RPN1210B, GE Healthcare). Membranes were blocked at room temperature for 1 hour in TBS-T (Tris-Buffered Saline (TBS), 0.05% Tween-20) containing 5% skim milk, and incubated with the anti-CPD antibody. Membranes were washed four times in TBS-T, then incubated for 1 hour with the appropriate peroxidase-conjugated anti-mouse secondary antibody at 1/5,000 (EMD Millipore). Immunoreactive proteins were visualized using the LumiGLO chemiluminescent substrate (Cell Signaling).

Luciferase assays

-1500 ~ +120 bp of ERCC1 promoter was cloned into the pGL3-basic reporter vector (Promega) to make a ERCC1-P-Luc luciferase reporter plasmid. For drug treatment reporter assay, 500 ng of reporter vector plus 50 ng of the renilla luciferase plasmid was transfected to cells using FuGENE 6 (Promega); cells were then treated with DMSO, 5 μ M GSK3368715 and 5 μ M GSK3235025 for 48 hours. For PRMT1 and PRMT5 CRISPR KO reporter assay, cells infected with control, PRMT1-sgRNA and PRMT5-sgRNA lentivirus were then transfected with reporter vector and renilla, and cells were harvested 48 hours after transfection. Reporter assays were performed using a dual luciferase reporter assay system (Promega) by Fluoroskan Ascent FL fluorometer (Thermo Fisher Scientific).

In vivo treatment experiment

For MDA-MB-231 xenograft experiments, three million tumor cells were injected subcutaneously into 6-weeks-old athymic female nu/nu mice (Stock No: 002019, Jackson Labs) and grown until tumors reached a size of approximately 30 mm³. Xenografted mice were randomized and then received vehicle, 50 mg/kg olaparib ip, 25 mg/kg GSK3368715 po, 50 mg/kg GSK3235025 po or the combination of two agents and three agents, 5 days a week for 3 weeks (n=6~8 per group). Caliper measurements were taken every three days starting from the initiation of drug treatment. Tumor volumes were calculated according to the formula: tumor volume [mm³] = (1/6) $\times$ π $\times$ (tumor length) $\times$ (tumor width)². Due to the nature of the performed experiments, no randomization and no blinding was used because it was deemed unfeasible. However, the resulting tumors were analyzed in a blinded manner. All animal procedures were in accordance with protocols approved by the Institutional Animal Care and Use Committee of the University of Pennsylvania.

Estimation of impact score of siRNA experiment

For a specific gene g , we used $\exp(t)_g$ to represent its expression measured by RNA-seq in a cell line treated with a given drug t , and $\exp C_g$ to represent its expression in a control treatment. Fold changes of expression were calculated for a specific gene by comparing its expression in each treatment condition to control treatment,

$$F_{\exp(t)_g} = \log_2(\exp(t)_g) - \log_2(\exp C_g).$$

For a specific gene g , we used $IC50(k)_g$ to represent IC50 of PARPi treatment in a cell line after knockdown this gene by pooled siRNA transfection, and $IC50C$ to represent IC50 of PARPi treatment in a control siRNA transfection. Fold changes of IC50 were calculated for a specific gene knockdown by comparing PARPi IC50 in the same cell line in siRNA knockdown condition to control siRNA transfection,

$$FIC50(k)_g = \frac{IC50(k)_g}{IC50C}.$$

Fold changes of IC50 were then normalized by min-max feature scaling,

$$F'IC50(k)_g = \frac{FIC50(k)_g - FIC50(k)_{\min}}{FIC50(k)_{\max} - FIC50(k)_{\min}}.$$

A final priority score was calculated for each gene examined,

$$Score_g = F'IC50(k)_g * \sum Fexp(t)_g.$$

TCGA RNA-seq data processing and gene expression analysis

The poly(A)+ RNA-seq data for primary tumors and their adjacent tissues were generated by the University of North Carolina and the British Columbia Cancer Agency Genome Sciences Centre as part of the TCGA project. All RNA-seq data were processed through a pipeline developed by the UCSC Toil RNAseq Recompute Compendium, which allowed us to consistently process large-scale RNA-seq data without computational batch effects¹⁰⁹. For TCGA RNA-seq data, if more than one sample existed for a participant, one single tumor sample (and matched adjacent sample, if applicable) was selected based on the following rules: (1) tumor sample type: primary (01) > recurrent (02) > metastatic (06); (2) order of sample portions: higher portion numbers were selected; and (3) order of plate: higher plate numbers were selected.

TCGA mRNA differential expression meta-analysis

For each cancer type which had at least three normal tumor-adjacent tissues in TCGA, differential gene expression analysis between tumor tissues and normal tumor-adjacent tissues was performed using one sided *t*-test (for up-regulation and down-regulation respectively). P values across all cancer types were then combined via Fisher's combined probability test¹¹⁰ to yield a meta-p value to assess the degree of up-regulation or down-regulation at the pan-cancer level. The meta-p value on the direction with the more significant dysregulation was reported for the corresponding PRMT genes.

CPTAC proteomic data processing and protein expression analysis

Extensive mass spectrometry-based proteomics data using isobaric tagging approaches (iTRAQ or TMT) for selected cancer types were generated by the National Cancer Institute's CPTAC. Protein-level processed data consisting of iTRAQ or TMT log(ratios) were downloaded from the CPTAC data portal (<https://cptac-data-portal.georgetown.edu>) on January 28, 2021.

CPTAC protein differential expression meta-analysis

Differential expression analysis between tumor tissues and paired normal tissues was performed using one sided *t*-test (for up-regulation and down-regulation respectively). P values across all cancer types were then combined via Fisher's combined probability test¹¹⁰ to yield a meta-p value to assess the degree of up-regulation or down-regulation at the pan-cancer level. The meta-p value on the direction with the more significant dysregulation was reported for the corresponding PRMT.

TCGA survival meta-analysis

For each PRMT gene in a given cancer type, we fit a Cox proportional hazards model and obtained a z-statistic and a p-value (with direction of effect) to assess whether high expression was associated with favorable or unfavorable overall survival. P-values across all cancer types were then combined via Fisher's combined probability test¹¹⁰ to yield a meta-p value to assess the prognostic value at pan-cancer levels (for risky effect and protective effect separately). The meta-p value on the direction with the more significant prognostic association was reported for the corresponding PRMT gene.

Recurrent genomic analysis

Recurrent somatic copy number alterations and mutations of the PRMT family were analyzed in the TCGA sample cohort by a standard pipeline developed by the Functional Cancer Genome project¹¹¹[111](#)–¹¹³[113](#). The G-score (somatic copy number alterations) and M-score (mutations) were established for each PRMT gene at both individual and pan-cancer levels.

Statistical analysis

Large-scale and multi-dimensional profiling data generated by the publicly accessible databases (TCGA, CPTAC, and DepMap) were used, therefore statistical analysis was not used to predetermine sample size in this study. For TCGA analysis, if more than one profiling file existed for a patient in TCGA, only one single file will be selected and used, and detailed methods for exclusion of duplicated profiling files are described in the method section. The computational analyses were not randomized, and the investigators were not blinded during data analyses of this study. When applicable, enrichment was tested using Fisher's exact test with FDR correction. Cell viability and gene relative expression data were shown as means with standard deviation (SD). Comparisons between groups were performed using student t-test.

Data and code availability

The genomic profiles of human cancers were generated by the TCGA project, which are publicly available through the Genomic Data Commons portal (GDC, <https://gdc-portal.nci.nih.gov>[111](#)). Protein-level processed data consisting of iTRAQ or TMT log(ratios) were downloaded from the CPTAC data portal (<https://cptac-data-portal.georgetown.edu>[112](#)). Genetic screening profiles in human cancer cell lines were generated by the DepMap and the Score projects, which are publicly available through the DepMap portal (<https://depmap.org/portal/>[113](#)), and the Score projects (<https://score.depmap.sanger.ac.uk/>[113](#)). The genomic data were retrieved, processed and analyzed through a master computational protocol developed by the Functional Cancer Genome project (FCG, <http://fcgportal.org/home/>[111](#)–¹¹³[113](#)) as described by our previous publications¹¹¹[111](#)–¹¹³[113](#) as well as the method section. This paper does not report original code. Any additional information required to reanalyze the data reported in this work paper is available from the corresponding authors upon request.

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Additional information

Author contributions

Y.Z., M.X., X.H., and L.Z. conceived and designed the study. J.Y., Z.H., and Y.W. performed computational, bioinformatics, and statistical analyses. Y.Z., M.X., J.J., Y.S., H.J., B.W., and J.S. conducted biological experiments. M.L., Y.F., K.T.M., J.L.T., S.H.K., R.H.V., and S.M.D. contributed to

the design of epigenetic inhibitor combinations in preclinical models and provided expertise in clinical oncology. O.T. and H.M.C. contributed to the design of epigenetic inhibitor combinations in preclinical models and provided expertise in early-stage drug development. Y.Z., M.X., X.H., and L.Z. wrote the manuscript.

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Additional files

Figure S1. The sensitivity of cancer cells to PARPi alone, PRMTi alone, and a PARPi and PRMTi combination in a panel of HGSOc and TNBC cell lines. For each treatment, the upper panel shows crystal violet staining of a colony formation assay; the lower left panel shows a quantified survival fraction; and the lower right panel shows the CI values. Fa, fraction affected. [↗](#)

Figure S2. Expression levels of the PRMTs across the TCGA tumor specimens. **A.** The heatmap of mRNA expression levels of PRMT family across the TCGA tumor samples. The intensity of heatmap color indicates the FPKM value for each percentile (25th, 50th, 75th, and 90th) of an individual PRMT gene in a given cancer type. **B.** The mRNA expression levels of the PRMT family in tumors compared with corresponding controls in the TCGA cohorts. Red and blue indicate up- and down-regulation, respectively. **C.** The protein expression levels of the PRMT family in tumors compared with corresponding controls in the CPTAC cohorts. Red and blue indicate up- and down-regulation, respectively. [↗](#)

Figure S3. Comet assay was used to measure DNA damage in cancer cell lines in which PRMTs were inhibited by chemical compounds or genetic approach. **A.** Left panels: A comet assay was used to measure DNA damage in OVCAR8, OVCAR3, MDA-MB-231, and MDA-MB-468 cell lines treated with DMSO, olaparib, GSK3368715, GSK3235025, or combinations. Scale bars, 10 μ m. Right panels: The extent of DNA damage was quantified using the tail moment in comet assays of OVCAR8, OVCAR3, MDA-MB-231, and MDA-MB-468 cell lines treated with DMSO, olaparib, GSK3368715, GSK3235025, or combinations. Data are presented as means $\pm$ SDs, * p < 0.05 determined by two-tailed Student's t tests. **B.** Left panels: A comet assay was used to measure olaparib treatment-induced DNA damage in OVCAR8, MDA-MB-231, and MDA-MB-468 cell lines in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Scale bars, 10 μ m. Right panels: The extent of olaparib treatment-induced DNA damage was quantified using the tail moment in comet assays of OVCAR8, MDA-MB-231, and MDA-MB-468 cell lines in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. Data are presented as means $\pm$ SDs, * p < 0.05 determined by two-tailed Student's t tests. [↗](#)

Figure S4. Repression of ERCC1 is a common mechanism by which PRMTi sensitize cells to PARPi treatment. **A.** The expression levels of aDNA and sDNA in cancer cell lines in which PRMTs were inhibited by chemical compounds. The expression levels of aDNA and sDNA in cancer in OVCAR8, OVCAR3, MDA-MB-231, and MDA-MB-468 cell lines treated with GSK3368715 or GSK3235025. **B.** The expression levels of aDNA and sDNA in cancer cell lines in which PRMTs were inhibited by genetic approach. The expression levels of aDNA and sDNA in cancer in OVCAR8 and MDA-MB-231 cell lines in which PRMT1 or PRMT5 was knocked out by CRISPR/Cas9. **C.** Dot blots of CPD levels in MDA-MB-231 cells in which ERCC1 was knocked out using lentiviral CRISPR/Cas9. Methylene blue staining was used as the loading control. [↗](#)

Figure S5. PRMT inhibition activates intrinsic innate immune pathways in cancer cells. **A.** The top 10 KEGG pathways in which genes upregulated by GSK3368715 (left) and GSK3235025 (right) treatments were significantly enriched. Circle size indicates fold enrichment. Lines indicate numbers of shared genes between pathways. The immune-related pathways are highlighted in red. **B.** Mosaic plots show percentages of genes upregulated by GSK3368715 (upper) and GSK3235025 (lower) treatments. Right: immune-related genes, left: non-immune-related genes. Significantly more immune-related genes,

compared with non-immune related genes, were upregulated by PRMT inhibition (GSK3368715: 17.8% vs 9.4%; GSK3235025: 23.6% vs 9.9%). **C.** Word cloud revealing the common immune-related genes upregulated by GSK3368715 and GSK3235025 treatments. 37.7% are associated with innate immune responses (dark red). Text size indicates meta-fold changes by both PRMTis. **D.** Scaled Venn diagram showing the functional families among the common upregulated immune-related genes. **E.** Western blot analysis of p-TBK1 expression in OVCAR8 (upper panel) and MDA-MB-231 (lower panel) cells treated with DMSO, olaparib (5 μ M), GSK3368715 (OVCAR8, 5 μ M and MDA-MB-231, 2.5 μ M), GSK3235025 (5 μ M) or olaparib (5 μ M) combined independently with GSK3368715 (OVCAR8, 5 μ M and MDA-MB-231, 2.5 μ M) and GSK3235025 (5 μ M) for 96 h. **F.** Western blot analysis of p-TBK1 expression in OVCAR8 (upper panel) and MDA-MB-231 (lower panel) cells in which PRMT1 and PRMT5 were independently knocked out using lentiviral CRISPR/Cas9. The cells were treated with olaparib (5 μ M, 96 h) or DMSO. **G.** Representative cytoplasmic micronuclei stained by PicoGreen in OVCAR8 (upper panel) and MDA-MB-231 (lower panel) cells treated with DMSO, olaparib (5 μ M), GSK3368715 (OVCAR8, 5 μ M and MDA-MB-231, 2.5 μ M), GSK3235025 (5 μ M), and olaparib (5 μ M) combined independently with GSK3368715 (OVCAR8, 5 μ M and MDA-MB-231, 2.5 μ M) and GSK3235025 (5 μ M). White arrows: micronuclei. Scale bar: 10 μ m. **H.** Quantification of micronuclei from G. Micronuclei number per cell was normalized to DMSO. Statistical significance was assessed using a two-tailed Student's *t* test. Error bars represent means $\pm$ SDs. [🔗](#)

Figure S6. PRMT1 and PRMT5 inhibition acts synergistically to enhance PARPi sensitivity in a panel of HGSOc and TNBC cell lines. [🔗](#)

Supplemental tables. **Table S1.** List of the epigenetic modulators that were used in initial drug combination screen. **Table S2.** List of the specimens that were used in this study from the TCGA project. **Table S3.** Expression of the PRMTs across cancers. **Table S4.** Recurrent genomic alterations of the PRMTs across cancers. **Table S5.** Cell dependencies of PRMTs from the DepMap project. **Table S6.** Correlations between the expression levels of PRMT1/5 and “50 hallmark” molecular signatures across the TCGA tumor specimens. **Table S7.** List of the potential BRCAness genes in the human genome. **Table S8.** List of the BRCAness genes that were repressed by PRMTi treatment. **Table S9.** PRMTi and PARPi combination concentration in each cancer line. **Table S10.** Primers and oligos that were used in this study. [🔗](#)

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Reviewer #2 (Public review):

Summary:

The authors show that a combination of arginine methyltransferase inhibitors synergize with PARP inhibitors to kill ovarian and triple negative cancer cell lines in vitro and in vivo using preclinical mouse models.

Strengths and weaknesses

The experiments are well-performed, convincing and have the appropriate controls (using inhibitors and genetic deletions) and use statistics.

They identify the DNA damage protein ERCC1 to be reduced in expression with PRMT inhibitors. As ERCC1 is known to be synthetic lethal with PARPi, this provides a mechanism for the synergy. They use cell lines only for their study in 2D as well as xenograph models.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors aimed to enhance the effectiveness of PARP inhibitors (PARPi) in treating high-grade serous ovarian cancer (HGSOC) and triple-negative breast cancer (TNBC) by inhibiting PRMT1/5 enzymes. They conducted a drug screen combining PARPi with 74 epigenetic modulators to identify promising combinations.

Zhang et al. reported that protein arginine methyltransferase (PRMT) 1/5 inhibition acts synergistically to enhance the sensitivity of Poly (ADP-ribose) polymerase inhibitors (PARPi) in high-grade serous ovarian cancer (HGSOC) and triple-negative breast cancer (TNBC) cells. The authors are the first to perform a drug screen by combining PARPi with 74 well-characterized epigenetic modulators that target five major classes of epigenetic enzymes. Their drug screen identified both PRMT1/5 inhibitors with high combination and clinical priority scores in PARPi treatment. Notably, PRMT1/5 inhibitors significantly enhance PARPi treatment-induced DNA damage in HR-proficient HGSOC and TNBC cells through enhanced maintenance of gene expression associated with DNA damage repair, BRCAness, and intrinsic innate immune pathways in cancer cells. Additionally, bioinformatic analysis of large-scale genomic and functional profiles from TCGA and DepMap further supports that PRMT1/5 are potential therapeutic targets in oncology, including HGSOC and TNBC. These results provide a strong rationale for the clinical application of a combination of PRMT and PARP inhibitors in patients with HR-proficient ovarian and breast cancer. Thus, this discovery has a high impact on developing novel therapeutic approaches to overcome resistance to PARPi in clinical cancer therapy. The data and presentation in this manuscript are straightforward and reliable.

Strengths:

(1) Innovative Approach: First to screen PARPi with a large panel of epigenetic modulators.

(2) Significant Results: Found that PRMT1/5 inhibitors significantly boost PARPi effectiveness in HR-proficient HGSOC and TNBC cells.

(3) Mechanistic Insights: Showed how PRMT1/5 inhibitors enhance DNA damage repair and immune pathways.

(4) Robust Data: Supported by extensive bioinformatic analysis from large genomic databases.

Weaknesses:

- (1) Novelty Clarification: Needs clearer comparison to existing studies showing similar effects.*
- (2) Unclear Mechanisms: More investigation is needed on how MYC targets correlate with PRMT1/5.*
- (3) Inconsistent Data: ERCC1 expression results varied across cell lines.*
- (4) Limited Immune Study: Using immunodeficient mice does not fully explore immune responses.*
- (5) Statistical Methods: Should use one-way ANOVA instead of a two-tailed Student's t-test for multiple comparisons.*

We sincerely thank Reviewer #1 for the insightful and constructive feedback, as well as for the kind acknowledgment of the significance of our work: “These results provide a strong rationale for the clinical application of a combination of PRMT and PARP inhibitors in patients with HR-proficient ovarian and breast cancer. Thus, this discovery has a high impact on developing novel therapeutic approaches to overcome resistance to PARPi in clinical cancer therapy. The data and presentation in this manuscript are straightforward and reliable.” We greatly appreciate the reviewer #1’s thoughtful comments, which have significantly improved the quality of our manuscript. In response, we conducted additional experiments and analyses, and made comprehensive revisions to the text, figures, and supplementary materials. In the “Recommendations for the authors” sections, we have provided point-by-point responses to each of the reviewer’s comments, which were immensely helpful in guiding our revisions. We believe these updates have substantially strengthened the manuscript and have fully addressed all reviewer concerns.

Reviewer #2 (Public Review):

Summary:

The authors show that a combination of arginine methyltransferase inhibitors synergize with PARP inhibitors to kill ovarian and triple-negative cancer cell lines in vitro and in vivo using preclinical mouse models.

PARP inhibitors have been the common targeted-therapy options to treat high-grade serous ovarian cancer (HGSOC) and triple-negative breast cancer (TNBC). PRMTs are oncological therapeutic targets and specific inhibitors have been developed. However, due to the insufficiency of PRMTi or PARPi single treatment for HGSOC and TNBC, designing novel combinations of existing inhibitors is necessary. In previous studies, the authors and others developed an “induced PARPi sensitivity by epigenetic modulation” strategy to target resistant tumors. In this study, the authors presented a triple combination of PRMT1i, PRMT5i and PARPi that synergistically kills TNBC cells. A drug screen and RNA-seq analysis were performed to indicate cancer cell growth dependency of PRMT1 and PRMT5, and their CRISPR/Cas9 knockout sensitizes cancer cells to PARPi treatment. It was shown that the cells accumulate DNA damage and have increased caspase 3/7 activity. RNA-seq analysis identified BRCAness genes, and the authors closely studied a top hit ERCC1 as a downregulated DNA damage protein in PRMT inhibitor treatments. ERCC1 is known to be synthetic lethal with PARP inhibitors. Thus, the authors add back ERCC1 and reduce the effects of PRMT inhibitors suggesting PRMT inhibitors mediate, in part, their effect via ERCC1 downregulation. The combination therapy (PRMT/PARP) is validated in 2D cultures of cell lines (OVCAR3, 8 and MDA-MB-231) and has shown to be effective in nude mice with MDA-MB-231 xenograph models.

Strengths and weaknesses:

Overall, the data is well-presented. The experiments are well-performed, convincing, and have the appropriate controls (using inhibitors and genetic deletions) and statistics.

They identify the DNA damage protein ERCC1 to be reduced in expression with PRMT inhibitors. As ERCC1 is known to be synthetic lethal with PARPi, this provides a mechanism for the synergy. They use cell lines only for their study in 2D as well as xenograph models.

We sincerely thank Reviewer #2 for the insightful and constructive feedback, as well as for the kind acknowledgment of the significance of our work: “Overall, the data are well-presented. The experiments are well-performed, convincing, and supported by appropriate controls (using inhibitors and genetic deletions) and statistics.” We greatly appreciate the reviewer #2’s thoughtful comments, which have significantly improved the quality of our manuscript. In response, we conducted additional experiments and analyses, and made comprehensive revisions to the text, figures, and supplementary materials. In the “Recommendations for the authors” sections, we have provided point-by-point responses to each of the reviewer’s comments, which were immensely helpful in guiding our revisions. We believe these updates have substantially strengthened the manuscript and have fully addressed all reviewer concerns.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Recent studies have revealed promising synergistic effects between PRMT inhibitors and chemotherapy, as well as DDR-targeting drugs (ref. 89-92). In the discussion, the authors should highlight what is novel in this study compared to the reported studies.

We thank the reviewer for this important comment and fully agree that prior studies have demonstrated the potential of PRMT inhibitors to enhance the efficacy of DNA damage-targeting agents and certain chemotherapies[1-4]. In response to the reviewer’s constructive suggestion, we have now revised the discussion to highlight the novel aspects of our study compared to previously reported findings. Specifically, our work presents several key advances that go beyond prior studies. Below, we would like to emphasize the novelty of our current study as follows:

In the clinic, a strategy termed “induced PARP inhibitor (PARPi) sensitivity by epigenetic modulation” is being evaluated to sensitize homologous recombination (HR)-proficient tumors to PARPi treatments. Together with other groups, we reported that repression of BET activity significantly reduces the expression levels of essential HR genes by inhibiting their super-enhancers[5]. This preclinical discovery is now being assessed in a Phase 1b/2 clinical trial combining the BET inhibitor ZEN-3694 with the PARPi talazoparib for the treatment of patients with metastatic triple-negative breast cancer (TNBC) who do not carry germline BRCA1/2 mutations. Promising anti-tumor activity has been observed in this ongoing trial[6]. Importantly, gene expression profiles from paired tumor biopsies demonstrated robust target engagement, evidenced by repression of BRCA1 and RAD51 mRNA expression, consistent with our preclinical findings in xenograft models. Based on these encouraging results, the trial is being expanded to a Phase 2b stage to enroll additional TNBC patients. Moreover, other combination strategies[7-13] based on this “induced PARPi sensitivity by epigenetic modulation” approach have also shown promising clinical responses in both intrinsic and acquired HR-proficient settings. Notably, these clinical studies indicate that the strategy is well-tolerated, likely due to cancer cells being particularly sensitive to epigenetic repression of DNA damage response (DDR) genes, compared with normal cells.

However, two key clinical challenges remain for broader application of this strategy in oncology: 1) which clinically actionable epigenetic drugs can produce the strongest synergistic effects with PARPi? and 2) can a BRCA-independent approach be developed? To address these questions, we performed a drug screen combining the FDA-approved PARPi olaparib with a panel of clinically relevant epigenetic drugs. This panel includes 74 well-characterized epigenetic modulators targeting five major classes of epigenetic enzymes, comprising 7 FDA-approved drugs, 14 agents in clinical trials, and 54 in preclinical development. Notably, both type I PRMT inhibitors (PRMTi) and PRMT5 inhibitors (PRMT5i) achieved high combination and clinical prioritization scores in the screen. Functional assays demonstrated that PRMT inhibition markedly enhances PARPi-induced DNA damage in HR-proficient cancer cell lines. In line with a strong positive correlation between PRMT and DDR gene expression across primary tumors, we observed that PRMT activity supports the transcription of DDR genes and maintains a BRCAness-like phenotype in cancer cells. These findings provide strong rationale for clinical development of PRMT/PARPi combinations in patients with HR-proficient ovarian or breast cancers. Mechanistic characterization from our study further supports PRMTi clinical development by elucidating mechanisms of action, identifying rational combinations, defining predictive biomarkers, and guiding dosing strategies.

We believe our studies will be of significant interest to the cancer research community for several reasons. First, they address major clinical challenges in women's cancers, specifically, high-grade serous ovarian cancer (HGSOC) and TNBC, both of which are aggressive malignancies with limited therapeutic options. Second, they offer a novel solution to overcome PARPi resistance. Our earlier discovery of "induced PARPi sensitivity by epigenetic modulation" has already shown promising clinical results and represents a new path to overcome both primary and acquired resistance to PARPi and platinum therapies. Third, they focus on a clinically translatable drug class. Selective and potent PRMT inhibitors have been developed by leading pharmaceutical companies, with more than ten currently in advanced clinical trials. Fourth, they support mechanism-driven combination strategies. Preclinical evaluation of PRMTi-based combinations with other therapeutic agents is urgently needed for future clinical success. Finally, our work highlights understudied but therapeutically relevant mechanisms in cancer biology. In-depth mechanistic analysis of the PRMT regulome is essential, and our studies provide important new insights into how PRMTs regulate transcription, RNA splicing, DNA damage repair, and anti-tumor immune responses in the context of HGSOC and TNBC.

In summary, our study identifies PRMT1 and PRMT5 as key epigenetic regulators of DNA damage repair and shows that their inhibition sensitizes HR-proficient tumors to PARP inhibitors by repressing transcription and altering splicing of BRCAness genes. Distinct from prior strategies, dual inhibition of type I PRMT and PRMT5 exhibits strong synergy, allowing for lower-dose combination treatments that may reduce toxicity. Our findings also nominate ERCC1 as a potential predictive biomarker and suggest that MYC-driven tumors may be particularly responsive to this approach. Collectively, these results offer a mechanistic rationale and translational framework to broaden the clinical application of PARP inhibitors.

(2) In Figures 3H-J, MYC targets were likely to correlate with the expression levels of PRMT1/PRMT5 in various public datasets, supporting previous reports that the Myc-PRMT loop plays critical roles during tumorigenesis (ref. 45). "Myc-targets" signatures were also the most significant signatures correlated with the expression of PRMT1 and PRMT5. The authors suggest that under MYC-hyperactivated conditions, tumors may be extremely sensitive to PRMT inhibitors or PRMTi/PARPi combination. However, the underlying mechanism remains unclear.

We sincerely thank the reviewer for the critical and insightful comments. We fully agree that more direct evidence is needed to establish the regulatory relationship between MYC and PRMT1/5. To investigate the effect of c-Myc on PRMT1 and PRMT5 expression, we analyzed RNA-seq data from P493-6 Burkitt lymphoma cells, which harbor a tetracycline (Tet)-repressible MYC transgene. In this system, MYC expression can be suppressed to very low levels and then reactivated, enabling a gradual increase in c-Myc protein levels[14]. Upon Tet removal to induce MYC expression, we observed a robust upregulation of both PRMT1 (4.3-fold) and PRMT5 (3.6-fold) RNA levels within 24 hours, as measured by RNA-seq. These findings indicate that MYC activation can transcriptionally upregulate PRMT1 and PRMT5. To determine whether this regulation is directly driven by MYC, we further analyzed MYC ChIP-seq profiles from the same cell line following 24 hours of MYC induction. Consistently, we observed remarkably increased MYC binding at the promoter regions of both PRMT1 and PRMT5 genes. Interestingly, MYC's regulatory influence was not limited to PRMT1 and PRMT5, we also observed transcriptional upregulation of other PRMT family members, including PRMT3, PRMT4, and PRMT6, in response to MYC activation. Together with the data presented in Figure 3H, these new results strongly suggest that MYC directly upregulates the expression of PRMT family genes by binding to their promoter regions. Consequently, increased PRMT expression may facilitate MYC's regulation of target gene expression and splicing in cancer cells. In cancers with MYC hyperactivation, this feed-forward loop may be amplified, creating a potential therapeutic vulnerability. In response to the reviewer's insightful suggestion, we have further explored how MYC regulates PRMT1/5 and whether this regulation modulates the efficacy of PRMT inhibitors in oncology. These unpublished observations are currently being prepared for a separate manuscript, and we have now incorporated a discussion of these unpublished findings into the revised version of this manuscript. We thank the reviewer again for the thoughtful and constructive comments regarding the MYC-PRMT regulatory axis.

(3) In Figure 5F, ERCC1 expression was unlikely to be reduced in cells treated with GSK025, especially in OVCAR8 cells, although other cells, including TNBC cells, are dramatically changed after treatment.

We sincerely thank the reviewer for the critical and insightful comments. We agree with the reviewer that in Figure 5F, although GSK025 treatment reduced ERCC1 expression, the loading control Tubulin also showed a notable decrease in the OVCAR8 cell line. This may be because Tubulin expression is not specifically affected by the chemical inhibitor GSK025 in this particular cell line, or it may be secondarily reduced as a consequence of PRMT inhibitor-induced cell death. As the reviewer pointed out, this phenomenon was not observed in other cell lines, suggesting that the effect on Tubulin is not specific to PRMT inhibition. To further investigate, we employed CRISPR/Cas9-mediated knockout of PRMT1 or PRMT5 in OVCAR8 cells, a more specific genetic approach to inhibit PRMT activity. In both cases, ERCC1 expression was significantly reduced, whereas Tubulin levels remained stable (Figure 5G). These results support the conclusion that PRMT1 and PRMT5 specifically regulate ERCC1 expression in OVCAR8 cells. The inconsistent effect on Tubulin is likely due to nonspecific cellular responses to chemical inhibition, which are generally more variable and less precise than those induced by genetic perturbation.

(4) In Figure 7H-L, MDA-MB-231 cells were implanted subcutaneously in nude immunodeficient mice to confirm the synergistic therapeutic action of the PRMTi/PARPi combination in vivo. Although PRMT inhibition activates intrinsic innate immune pathways in cancer cells, suggesting that PRMTi treatments may enhance intrinsic immune reactions in tumor cells, the use of nude immune deficient mice means that changes in the tumor immune microenvironment remain unknown.

We sincerely thank the reviewer for the critical and insightful comments. We fully agree with the reviewer that our *in vivo* experiments using the human cancer cell line MDA-MB-231 in immunodeficient nude mice limit our ability to assess changes in the tumor immune microenvironment. We thank the reviewer for highlighting this important limitation. While the primary goal of the current study was to investigate the therapeutic synergy between PRMT inhibition and PARP inhibition in cancer cells, we would like to take this opportunity to share additional unpublished data that further support and extend the reviewer's point regarding the immunomodulatory effects of PRMT inhibitors. In syngeneic mouse tumor models, we have observed that the combination of PRMT inhibition and PARP inhibition leads to a more robust anti-tumor immune response compared to either treatment alone. Specifically, we found increased infiltration of CD8⁺ cytotoxic T cells within the tumor microenvironment, suggesting enhanced immune activation and tumor immunogenicity. Furthermore, we have also obtained preliminary evidence that PRMT inhibition can potentiate immune checkpoint blockade therapy. Mechanistically, this may be mediated through the activation of the STING1 pathway and the upregulation of splicing-derived neoantigens, both of which have been implicated in promoting tumor immune visibility. These findings indicate that beyond enhancing DNA damage response, PRMT inhibition may have a broader impact on tumor-immune interactions and could serve as a promising strategy to sensitize tumors to immunotherapy. A separate manuscript detailing these results is currently in preparation and will be submitted for publication as an independent research article. In light of the reviewer's thoughtful suggestions and in consideration of feedback from Reviewer #2, who recommended removing Figure 6 from the manuscript, we have carefully reevaluated the overall organization of the manuscript. Given the scope and focus of the current work, as well as the desire to maintain a concise and coherent narrative, we decided to move the content originally presented in Figure 6 to the supplementary materials. This figure is now included as Supplementary Figure S5 in the revised version of the manuscript. We believe this change helps streamline the main text while still making the additional data available for interested readers.

(5) In Figures 6-7, a two-tailed Student's t-test was used to determine the statistical differences among multiple comparisons, which should be performed by one-way ANOVA followed by a post hoc test.

We thank the reviewer for this thoughtful and important comment regarding the choice of statistical method. We fully agree with the reviewer that one-way ANOVA followed by a post hoc test is one of the standard approaches for multiple group comparisons. In response to the suggestion, we have performed one-way ANOVA on our data and found that the statistical conclusions are consistent with those obtained from the two-tailed Student's t-tests. For example, in the first panel of Figure 6A (OVCAR8 treated with GSK715), one-way ANOVA ($p = 1.1 \times 10^{-6}$), followed by Tukey's HSD test, confirmed significant differences between control and Olaparib ($p = 0.000165$), control and GSK715 ($p = 0.000145$), control and combination ($p = 6.067 \times 10^{-7}$), Olaparib and combination ($p = 0.0003523$), and GSK715 and combination ($p = 0.0004015$), consistent with the conclusions from the two-tailed t-test shown in Figure 6H. Additionally, we would like to explain why two-tailed Student's t-tests were used in our current study. When comparisons are predefined and conducted pairwise (i.e., two groups at a time), a two-tailed Student's t-test is statistically equivalent to one-way ANOVA for those comparisons. In our study, each comparison involved only two groups, and we therefore chose t-tests for hypothesis-driven, specific comparisons rather than exploratory multiple testing. This approach aligns with valid statistical principles. All statistical analyses presented in Figures 6-7 were designed to evaluate specific, biologically meaningful comparisons (e.g., treatment vs. control or treatment A vs treatment B). The study was hypothesis-driven, not exploratory, and did not involve simultaneous comparisons across multiple groups. In such cases, the t-test provides a more direct and interpretable result for targeted comparisons. The

use of Student's t-tests reflects the focused nature of the analysis, where each test directly addresses a specific biological question rather than a global group comparison. We sincerely appreciate the reviewer's thoughtful comments on the statistical methods.

Reviewer #2 (Recommendations for the authors):

(1) If the authors kept the tumors of various sizes in Figure 7I, it would be important to assess the protein and/or mRNA level of ERCC1 to further support their mechanism.

We sincerely thank the reviewer for the insightful comments. We fully agree that evaluating ERCC1 expression in drug-treated tumor samples is critical to support the proposed mechanism. Due to the limited volume of tumor specimens and extensive necrosis observed after three weeks of treatment in the condition used for Figure 7I, we were unable to obtain sufficient material for expression analysis in the original cohort. To address this, we conducted an additional experiment using xenograft-bearing mice (MDA-MB-231 model), initiating treatment when tumors reached approximately 200 mm³ to ensure adequate tissue collection. We also shortened the treatment duration to 7 days to assess early molecular responses to therapy, rather than downstream effects. Consistent with our in vitro results, both GSK715 and GSK025 significantly reduced ERCC1 RNA expression (0.79 ± 0.17 , $p = 0.03$; 0.82 ± 0.11 , $p = 0.02$, respectively), and the combination treatment further decreased ERCC1 expression (0.49 ± 0.20 , $p = 0.0003$), as determined by qRT-PCR. A two-tailed Student's t-test was used for statistical analysis. In this experiment, we used the same dosing regimen as in the three-week treatment shown in Figure 7I. Importantly, the shorter treatment period and moderate tumor size at treatment initiation minimized necrosis and did not significantly affect tumor growth, allowing for reliable molecular evaluation. We sincerely thank the reviewer for highlighting this important point.

(2) Figure 2G: please explain why two bands remain for sgPRMT1.

We greatly appreciate the reviewer for raising this insightful and important question. As the reviewer pointed out, an additional band appeared after PRMT1 knockdown in OVCAR8 cells using two sequence-independent gRNAs. Notably, this band was not observed in MDA-MB-231 cells. The antibody used to detect PRMT1 (clone A33, #2449, Cell Signaling Technology) is widely adopted in PRMT1 research, with over 65 citations supporting its specificity. Interestingly, previous studies[15] have identified seven PRMT1 isoforms (v1–v7), generated through alternative splicing and exhibiting tissue-specific expression patterns. Of these, three isoforms are detectable using the A33 antibody. We believe the additional band observed upon sgRNA treatment likely represents a PRMT1 isoform that is normally expressed at low levels in OVCAR8 cells. Upon knockdown of the major isoforms by CRISPR/Cas9, expression of this minor isoform may have increased as part of a compensatory feedback mechanism, rendering it detectable by immunoblotting. Because PRMT1 isoform expression is largely tissue-type specific, it is not surprising that the same band was absent in MDA-MB-231 cells, which are derived from a different lineage than OVCAR8 cells. The reviewer raised an important question regarding the role of PRMT1 isoforms in regulating DNA damage response in cancer. We agree this is an intriguing direction and will investigate it further in future studies.

(3) Figure 4D: Please correct the figure legend so the description matches the color in the figure. Red and blue are absent.

We sincerely thank the reviewer for the critical and insightful comments. The figure legend for Figure 4D has been corrected in the revised version of the manuscript to accurately match the colors shown in the figure. We thank the reviewer for pointing out this issue.

(4) *Figure 7A and B: please indicate the cell lines used.*

We sincerely thank the reviewer for the critical and insightful comments. In Figure 7A and 7B, human embryonic kidney 293T (HEK293T) cells were used due to their high transfection efficiency and widespread application in reporter assays. This information has been incorporated into the figure legend for Figures 7A and 7B.

(5) *What is the link with ERCC1 splicing because reduced overall ERCC1 expression is clear?*

We sincerely thank the reviewer for the critical and insightful comments. As the reviewer pointed out, although the direct impact of ERCC1 alternative splicing on its protein expression remains to be fully elucidated, it is likely that PRMT inhibition induces aberrant splicing events that result in the production of alternative ERCC1 isoforms with impaired or altered function. These splicing changes may compromise ERCC1's role in DNA repair pathways. Furthermore, as shown in Figure 4G, we observed a reduction in the total ERCC1 mRNA reads following PRMTi treatment. This decrease may be attributed, at least in part, to the instability of the alternatively spliced ERCC1 transcripts, which could be more prone to degradation. In combination with the transcriptional downregulation of ERCC1 induced by PRMT inhibition, these alternative splicing events may lead to a further reduction in functional ERCC1 protein levels. This dual impact on ERCC1 expression, through both decreased transcription and the generation of unstable or non-functional isoforms, likely contributes to the enhanced cellular sensitivity to PARP inhibitors observed in our study. We believe this represents an important mechanistic insight into how PRMT inhibition modulates the DNA damage response in cancer cells, and further studies are warranted to investigate the precise role of ERCC1 splicing regulation in this context. We thank the reviewer for pointing out this interesting future research direction.

(6) *Figure 7J: From the graph, it seems like Olaparib+G715 and G715+G025 have a similar effect on tumor volume (two curves overlap). Please discuss.*

We sincerely thank the reviewer for the critical and insightful comments. In the current study, the doses used for single-agent treatments were selected based on prior publications. For example, the dose of GSK715 was guided by a recent study from the GSK group[16]. Our in vitro and in vivo findings, together with previously published data, consistently demonstrate that GSK715 is more potent than both GSK025 and Olaparib. Notably, treatment with GSK715 alone led to significantly greater inhibition of tumor growth compared to either GSK025 or Olaparib administered individually. This higher potency of GSK715 also explains the comparable levels of tumor suppression observed in the combination groups, including GSK715 plus Olaparib and GSK715 plus GSK025. These results suggest that GSK715 is likely the primary driver of efficacy in the two drug combination settings. Importantly, this observation provides a valuable opportunity to further refine and optimize the dosing strategy for GSK715. Specifically, because GSK715 is highly potent, its dose may be reduced when used in combination regimens without compromising therapeutic efficacy. This approach could significantly improve the safety profile of GSK715 by minimizing potential dose-related toxicities, thereby enhancing its suitability for future clinical development in combination therapy contexts.

(7) *Discussion: "PRMT5i increased global sDMA levels"-> "... aDMA levels."*

We sincerely thank the reviewer for the critical and insightful comments. In response, we have corrected the sentence in the discussion from "PRMT5i increased global sDMA levels, which suggested that type I PRMT and PRMT5 share a substrate (i.e., MMA) and/or their

functions are compensatory” to “PRMT1i increased global sDMA levels, which suggested that type I PRMT and PRMT5 share a substrate (i.e., MMA) and/or their functions are compensatory.” We apologize for the misstatement and have corrected this error in the revised version of the manuscript.

(8) In addition to the methods, add that nude mice were used in the body of the results and the figure legend for Figure 7J.

We sincerely thank the reviewer for the critical and insightful comments. In the revised version of the manuscript, we have added that immunodeficient nude mice were used in both the body of the Results section and the figure legend for Figure 7J, in addition to the Methods section. We thank the reviewer for this helpful suggestion.

(9) Figure 6 can be deleted to focus the manuscript. It does not add to the PARP inhibition story, but only suggests a link to immunotherapy where this has been reported previously PMID: 35578032 and 32641491.

We sincerely thank the reviewer for the critical and insightful comments. Reviewer #1 also raised a related concern regarding the relevance of this section to the main focus of the manuscript. In consideration of both reviewers’ comments, we have decided to move the data previously shown in Figure 6 to the supplementary section as Supplementary Figure S5. This revision allows us to streamline the main text and maintain a clear focus on the core findings related to PARP inhibition. At the same time, we believe the immunotherapy-related observation may still be of interest to some readers. By presenting these results in the supplementary materials, we ensure that this potentially relevant link remains accessible without distracting from the primary narrative of the manuscript. We greatly appreciate the reviewers’ guidance in helping us improve the clarity and focus of our work. We thank the reviewer for the thoughtful suggestion.

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